INTERNATIONAL CYBERSECURITY AND DIGITAL FORENSICS ACADEMY

SBT-DF204: COMPUTER FORENSICS CASE STUDIES

CASE STUDY 3: INVESTIGATING WINDOWS MEMORY EVIDENCE

FINAL FORENSIC INVESTIGATION REPORT

Prepared by: Charles Uke
Student ID: 2025/FWSD/11466
Course: SBT-DF204 – Computer Forensics Case Studies
Analysis Platform: Kali Linux running in VMware
Primary Forensic Tool: Volatility Framework 2.6.1
Evidence Examined: `memdumpWin7.mem`
Report Date: 16 October 2026

Investigating Memory Evidence

DOCUMENT CONTROL

Field	Details
Case reference	SBT-DF204 Case Study 3
Examiner/student	Charles Uke
Student ID	2025/FWSD/11466
Primary evidence	Windows memory image, `memdumpWin7.mem`
Working directory	`~/SBT-DF204-CaseStudy3`
Operating system used for analysis	Kali Linux
Memory-analysis framework	Volatility Framework 2.6.1
Selected memory profile	`Win7SP1x86`
Evidence integrity method	SHA-256 cryptographic hashing
Report status	Final report based on recorded analysis outputs

1. EXECUTIVE SUMMARY

This investigation examined a supplied Windows memory image named `memdumpWin7.mem` using the Volatility Framework 2.6.1 in a Kali Linux virtual machine. The objective was to preserve the evidence, identify the likely operating system profile, recover relevant process and network information, inspect command history, and examine registry artifacts relating to user activity and removable storage.

The original image and working copy were verified using SHA-256 hashing. Both produced the same digest:

`500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e`

Both files were recorded as 1,073,676,288 bytes and had read only permissions at the time of the final verification. These results support the integrity of the working copy relative to the original file examined.

The image was consistent with a Windows 7 Enterprise SP1, 32-bit environment. The `Win7SP1x86` profile was selected for the principal Volatility queries after reviewing the `imageinfo` output.

The most significant finding was recovered console history containing the following command:

`copy secret_file.docx F:`

The console output also contained:

`1 file(s) copied.`

Separately, registry artifacts identified a device described as `General UDisk USB Device` and associated drive letter `F:` with a USB storage device path. These findings are consistent with a command line file copy operation targeting removable storage.

However, the memory evidence does not independently establish the contents of the file, the identity of the person who performed the operation, whether the action was authorised, or whether the destination file remained present after the recorded activity.

Other findings included a host IP address of `192.168.56.8`, two closed TCP connection records involving `192.168.56.5:80`, an SSH service process, multiple Chrome processes, an FTK Imager process, user profile registry artifacts, and UserAssist entries containing application and script related paths.

The investigation encountered limitations. Some Volatility plugins could not run because required dependencies were missing. The `psscan` output contained no process records, and some process timestamps were inconsistent with the image timestamp. These limitations are documented in the relevant sections.

Overall conclusion: The strongest finding is the correlation between recovered console history showing a file copy command to `F:` and registry evidence associating `F:` with a General UDisk USB storage device. This supports a cautious inference that the system retained evidence consistent with copying `secret_file.docx` to a removable storage volume. It does not, by itself, prove data exfiltration, malicious intent, or the identity of the person responsible.

2. OBJECTIVES

The investigation was conducted to achieve the following objectives:

1. Preserve the supplied memory image and verify the integrity of the working copy.
2. Identify the likely operating system profile and select a suitable Volatility profile.
3. Examine running process information and process scan results.
4. Recover available network artifacts and console command history.
5. Identify registry hives and examine relevant Windows registry information.
6. Investigate removable storage artifacts and drive letter mappings.
7. Examine selected user profile, RecentDocs, MountPoints2, and UserAssist artifacts.
8. Correlate significant observations into a UTC timeline.
9. Document limitations, uncertainty, and recommendations for further investigation.

3. SCOPE AND EVIDENCE HANDLING

3.1 Scope

The examination focused on the supplied Windows memory image and the command outputs generated during the laboratory workflow.

The analysis included:

- Memory image identification and profile selection.

- Process listing and scanning.
- Network artifact extraction.
- Console and command history recovery.
- Registry hive identification.
- USB storage and mounted volume analysis.
- Windows version and user profile examination.
- Selected RecentDocs and UserAssist registry analysis.
- Evidence hashing and output file inventory.

The investigation did not include a complete disk image, independent examination of the physical USB device, or independent recovery and validation of the destination copy of `secret_file.docx`.

3.2 Evidence identification

Item	Recorded information
Primary source image	`/home/kali/Desktop/Investigating Memory Evidence/memdumpWin7.mem`
Additional source copy observed	`/home/kali/Downloads/memdumpWin7.mem`
Working copy	`~/SBT-DF204-CaseStudy3/working/memdumpWin7.mem`
File size	1,073,676,288 bytes
Approximate size	1.0 GB
File identification	`data`
Source modification timestamp	`2026-10-16 16:33:14.816131616 -0400`
Analysis copy permissions	`-r--r--r--`
Source image SHA-256	`500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e`
Working copy SHA-256	`500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e`

3.3 Evidence preservation

The memory image was copied into the case workspace, and analysis was performed against the working copy rather than intentionally modifying the source image.

The final integrity check recorded identical SHA-256 digests for the original and working files. Their reported sizes and permissions also matched.

These checks support the conclusion that the working copy was identical to the source at the time of verification. They do not independently establish the source image's provenance or authenticity before it was received.

The workspace contained separate directories for original evidence, working evidence, outputs, evidence records, screenshots, notes, and the report.

4. TOOLS AND METHODOLOGY

4.1 Tools used

Tool	Purpose
Kali Linux	Controlled environment for memory analysis
Volatility Framework 2.6.1	Memory-image analysis and artifact extraction
Python 2.7.18	Runtime used by Volatility 2
`sha256sum`	Cryptographic integrity verification
`stat`	File size and permissions verification
`file`	File-type identification

| `find` | Output and evidence inventory |
| Windows Snipping Tool | Capture of screenshots for the evidence appendix |

The Volatility repository revision recorded during the workflow was:

`328a178edeecaf8d2abf96618ecfcb14a7b0c860`

4.2 Examination procedure

The investigation followed these general stages:

1. Identify the source image and record file metadata.
2. Create and verify the analysis copy.
3. Confirm the Volatility installation and identify the memory profile.
4. Run process, network, and console analysis plugins.
5. Enumerate registry hives and inspect relevant registry keys.
6. Correlate removable storage registry artifacts with console history.
7. Examine selected user profile artifacts.
8. Record findings, limitations, and evidence hashes.
9. Capture screenshots of relevant commands and results.

Commands and outputs were retained in the case workspace where available.

5. MEMORY IMAGE IDENTIFICATION

5.1 Profile identification

The Volatility `imageinfo` command suggested multiple possible profiles, including:

- `Win7SP1x86_23418`
- `Win7SP0x86`
- `Win7SP1x86_24000`
- `Win7SP1x86`

The principal analysis commands used `Win7SP1x86`.

The recorded image information included:

Field	Recorded result
Selected profile	`Win7SP1x86`
Kernel address-space indication	`IA32PagedMemoryPae`
Backing file	`working/memdumpWin7.mem`
DTB	`0x185000L`
KDBG	`0x82972c68L`
KPCR	`0x82973d00L`
CPU count	1
KUSER_SHARED_DATA	`0xffdf0000L`
Image timestamp	`2019-01-06 15:09:06 UTC`
Local time displayed by imageinfo	`2019-01-06 07:09:06 -0800`

A KDBG scan was also run and saved to `outputs/04_kdbgscan.txt`.

5.2 Operating-system identification

The Windows version registry output identified the operating system as:

- Windows 7 Enterprise.
- Version 6.1.
- Build 7601.
- Service Pack 1.

The selected profile was consistent with a 32-bit Windows 7 SP1 memory image.

The profile was treated as the best fit profile based on the available evidence and successful execution of several plugins. It should not be regarded as an absolute guarantee that every structure in the memory image was interpreted correctly.

6. PROCESS EXAMINATION

6.1 Process listing

The `pslist` plugin returned process records from the memory image. The principal output was saved as:

`outputs/02_pslist_Win7SP1x86.txt`

An additional process listing was saved as:

`outputs/21_pslist.txt`

Notable records included:

Process	PID	Recorded start time (UTC)	Observation
`sshd.exe`	2016	2019-01-06 15:02:49	SSH service process present
Chrome processes	2388, 1280, 2380, 2912, 976	Approximately 15:04:30–15:06:23	Multiple browser processes present
`cmd.exe`	3996	Approximately 15:06:11	Consistent with a command-line session
`conhost.exe`	3988	Approximately 15:06:11	Windows console host process present
`FTK Imager.exe`	2596	2019-01-06 15:07:15	Forensic imaging application represented in the process list
`dllhost.exe`	2228	2019-01-06 15:09:07	Process timestamp occurs shortly after the image timestamp

The process listing provides context for the system state represented in the memory image. Process presence alone does not establish the purpose of an activity or the identity of the person responsible.

6.2 Process scanning

The `psscan` plugin was run using the selected profile. The saved output contained no process records.

An empty `psscan` result does not prove that no hidden, terminated, or unlinked processes existed. The result may reflect limitations in the image, profile interpretation, plugin behaviour, or the memory structures available to the scanner.

The result is therefore documented as a limitation rather than evidence that no additional processes were present.

6.3 Timestamp anomalies

Two timestamp issues require caution:

1. `dllhost.exe` was reported as starting at `2019-01-06 15:09:07 UTC`, approximately one second after the image timestamp of `15:09:06 UTC`.
2. The System process, PID 4, was reported with a start time of `18:02:44 UTC`, while several other system and service records appeared around `15:02 UTC`.

The cause of these inconsistencies was not established. The timestamps are preserved as reported and should not be silently corrected or treated as a definitive event sequence.

7. NETWORK ARTIFACT EXAMINATION

The network analysis output was saved as:

`outputs/05_netscan.txt`

7.1 Host address

The recovered console output and network scan both recorded the host IPv4 address:

`192.168.56.8`

The console output also reported a subnet mask of `255.255.255.0` and a blank default gateway field.

7.2 TCP connection records

Two closed TCP records were identified:

Local endpoint	Remote endpoint	State	PID
`192.168.56.8:49177`	`192.168.56.5:80`	CLOSED	-1
`192.168.56.8:49178`	`192.168.56.5:80`	CLOSED	-1

These records establish that corresponding connection structures were recovered from memory. They do not establish that a successful transfer occurred, what data may have been transferred, or why the connections existed.

The PID value of `-1` also means the output does not associate these records with a specific process in the recovered results.

7.3 Chrome UDP endpoint

The network output included a Chrome associated UDP endpoint at:

`192.168.56.8:56647`

The recorded creation time was:

`2019-01-06 15:08:32 UTC`

This provides evidence of a network endpoint structure associated with Chrome in the captured system state. The destination and purpose of the associated activity were not established from the available output.

7.4 SSH service

The network output recorded `sshd.exe`, PID 2016, listening on:

`0.0.0.0:22`

This indicates an SSH service listening on the system's network interfaces in the captured state. The listening socket alone does not prove that an incoming connection occurred or that a remote user accessed the system.

8. CONSOLE HISTORY AND FILE-COPY ARTIFACT

8.1 Recovered command history

The command history output was saved as:

`outputs/06_cmdscan.txt`

The console output was saved as:

`outputs/07_consoles.txt`

The recovered console transcript contained the following sequence:

```text
Microsoft Windows [Version 6.1.7601]
C:\Users\IEUser>ipconfig
...
IPv4 Address: 192.168.56.8
Subnet Mask: 255.255.255.0
Default Gateway: [blank]
C:\Users\IEUser>cd Downloads
C:\Users\IEUser\Downloads>copy secret_file.docx F:
      1 file(s) copied.
```

The recovered command history identified the following commands:

```text
ipconfig
cd Downloads
copy secret_file.docx F:
```

8.2 Interpretation of the copy command

The command:

```text
copy secret_file.docx F:
```

attempts to copy a file named `secret_file.docx` from the current directory to the destination addressed by `F:`.

The console output also contained:

```text
1 file(s) copied.
```

The success text supports the interpretation that the recorded console session reported a successful copy operation.

This is a significant finding because the console command is corroborated by separate registry evidence associating `F:` with a USB storage device.

However, the memory artifact does not independently establish:

- The actual contents of `secret_file.docx`.
- Whether the source file was sensitive or confidential.
- Whether the destination file remained intact or present after the operation.
- The identity of the person who entered the command.
- Whether the operation was authorised.
- Whether the device was subsequently removed or used elsewhere.

The finding should therefore be described as evidence consistent with a file copy operation to removable storage, rather than conclusive proof of data theft or exfiltration.

9. REGISTRY AND REMOVABLE-STORAGE EXAMINATION

9.1 Registry hive enumeration

The registry hive list was saved as:

`outputs/08_hivelist.txt`

The output identified system hives including SYSTEM, SOFTWARE, SAM, SECURITY, DEFAULT, BCD, and Syscache, together with user-associated hives for IEUser and `sshd_server`.

The presence of these hives enabled further examination of system configuration, removable storage, and selected user profile artifacts.

9.2 Current control set

The SYSTEM hive's `Select` key was examined. The recorded values were:

Value	Result
Current	1
Default	1
Failed	0
LastKnownGood	2

These values indicate that ControlSet001 was the current control set and ControlSet002 was the last-known-good control set for the analysis performed.

The output was saved as:

`outputs/09_system_select.txt`

9.3 USBSTOR artifacts

The USBSTOR registry queries identified a device with the following identifier:

`Disk&Ven_General&Prod_UDisk&Rev_5.00`

The device was described as:

`General UDisk USB Device`

The relevant USBSTOR key was recorded as last updated at:

`2019-01-06 15:02:54 UTC`

The USBSTOR entry was found in both ControlSet001 and ControlSet002.

The device instance examined was:

```text
ControlSet001\Enum\USBSTOR\
Disk&Ven_General&Prod_UDisk&Rev_5.00\
6&1bec0f48&0&_&0
```

Relevant recorded details included:

Field	Recorded value
Device description	Disk drive
Friendly name	General UDisk USB Device
Class	DiskDrive
Service	disk
USBSTOR identifier	`Disk&Ven_General&Prod_UDisk&Rev_5.00`
Last-updated timestamp	`2019-01-06 15:02:54 UTC`

These registry entries support the presence of a USB storage device configuration in the examined Windows registry artifacts. They do not independently identify the physical owner of the device.

9.4 MountedDevices and F: drive

The MountedDevices output was saved as:

`outputs/12_mounted_devices.txt`

The registry data associated the drive letter `F:` with a USBSTOR-style device path. The associated volume GUID was:

`{421e7e52-11dd-11e9-b3cf-08002709e15d}`

The registry examination therefore identified a relationship between `F:` and a USB storage volume.

This is significant because the recovered console history contains the command:

```text
copy secret_file.docx F:
```

The relationship between the console history and the registry evidence strengthens the inference that the command targeted a removable storage volume.

The analysis also identified a USB device ContainerID:

`{421e7e37-11dd-11e9-b3cf-806e6f6e6963}`

This ContainerID is different from the volume GUID above. The two identifiers must not be treated as interchangeable.

9.5 MountPoints2 artifacts

The IEUser `MountPoints2` registry key was recorded as last updated at approximately:

`2019-01-06 15:03:07 UTC`

The key included entries for `F:` and the volume GUID:

`{421e7e52-11dd-11e9-b3cf-08002709e15d}`

The `MountPoints2\F` key was recorded as updated at approximately:

`2019-01-06 15:03:06 UTC`

These artifacts are consistent with the user's registry retaining information about a mounted volume. They provide corroborating context but should not be interpreted as a precise record of the exact physical insertion or removal time.

9.6 Removable-storage finding

The evidence can be summarised as follows:

1. USBSTOR registry entries identify a General UDisk USB storage device.
2. MountedDevices associates `F:` with a USB storage path.
3. MountPoints2 contains related `F:` and volume-GUID entries.
4. Recovered console history records `copy secret_file.docx F:`.
5. The console output contains the message `1 file(s) copied.`

Finding: These independent artifacts are consistent with a file named `secret_file.docx` being copied from the Downloads directory to a volume associated with a General UDisk USB storage device.

The evidence does not independently establish whether the activity was authorised, whether the file was confidential, or whether the data was later transferred to another system.

10. WINDOWS VERSION AND USER PROFILES

10.1 Windows version

The Windows version output was saved as:

`outputs/15_windows_version.txt`

The recorded operating-system information was:

Field	Result
Product	Windows 7 Enterprise
Version	6.1
Build	7601
Service pack	Service Pack 1
Installation type	Client
Architecture indication	Consistent with the selected x86 profile

The operating-system registry key was recorded as last updated at approximately `2019-01-06 14:32:50 UTC`.

10.2 User profiles

The ProfileList examination identified several SIDs, including entries ending in `-1000` and `-1002`.

The SID ending in `-1000` was associated with:

`C:\Users\IEUser`

The profile registry key was recorded as last updated at:

`2019-01-06 15:07:15 UTC`

The SID ending in `-1002` was associated with:

`C:\Users\sshd_server`

Its profile key had an earlier recorded timestamp of:

`2015-09-21 09:51:49 UTC`

These records establish the profile paths present in the registry artifacts. They do not establish the real-world identity of the person using either profile.

11. RECENTDOCS AND USERACTIVITY ARTIFACTS

11.1 RecentDocs

The IEUser RecentDocs key was recorded as last updated at:

`2019-01-06 14:32:14 UTC`

Recovered entries included script-related names such as:

- `preprovisioner.ps1`
- `OPENSSH.PS1`
- `post-win-updates.ps1`

Other entries included standard Windows items such as `Readme.txt`, `eula.txt`, `System32`, and Control Panel related entries.

The presence of a name in RecentDocs is evidence of a registry artifact referencing that item. It does not, by itself, prove that a script was executed.

The corresponding output was saved as:

`outputs/26_ieuser_recentdocs.txt`

11.2 Explorer registry key

The IEUser Explorer key was recorded as last updated at approximately:

`2019-01-06 15:02:54 UTC`

Its subkeys included standard Explorer related entries, among them:

- `RecentDocs`
- `RunMRU`
- `MountPoints2`
- `Shell Folders`
- `User Shell Folders`
- `UserAssist`
- `FileExts`

The presence of these keys provided locations for further user activity queries.

11.3 Run, TypedPaths, and RunMRU

The IEUser Run key query returned no values in the saved output.

The TypedPaths and RunMRU queries also returned no recovered values in the recorded outputs.

These results are limited to the queried keys and recovered data. They should not be interpreted as proof that the user never launched applications, entered paths, or performed other activities.

The requested Explorer key for the `sshd_server` hive was not found in the query output. This is recorded as a query result, not proof that the account had no activity.

11.4 UserAssist artifacts

The UserAssist Count key output was saved as:

`outputs/31_userassist_count_guid1.txt`

The key was recorded as last updated at:

`2019-01-06 15:08:42 UTC`

UserAssist registry value names are commonly ROT13-obfuscated. Manual decoding of selected strings identified the following labels:

Decoded artifact	Interpretation
`C:\Users\IEUser\Desktop\compact.bat`	A batch-file path recorded in UserAssist data
`C:\Users\IEUser\Desktop\AccessData FTK Imager3-2-0.exe`	A path referencing an FTK Imager executable
`C:\OTinfo\OTINFO.EXE`	A recorded application path
`D:\WINDOWSADDITIONS-K86.EXE`	A recorded executable path
`Microsoft.Windows.GettingStarted`	Windows component label
`Microsoft.Windows.RemoteDesktop`	Windows component label

These decoded strings may indicate recorded application interaction history. They are not sufficient by themselves to prove that every referenced executable successfully ran or that any referenced program was malicious.

The dedicated UserAssist plugin could not be used because the required `Crypto.Hash` dependency was unavailable. The raw registry output was obtained using a direct registry key query, and selected strings were decoded manually.

12. CONSOLIDATED UTC TIMELINE

The following timeline consolidates the significant timestamps observed in the saved outputs.

Timestamp (UTC)	Artifact	Significance
2019-01-06 14:32:14	IEUser RecentDocs key	Recent-document registry timestamp
2019-01-06 14:32:50	Windows version key	Operating-system registry timestamp
2019-01-06 15:02:49	`sshd.exe`, PID 2016	SSH service process start time
2019-01-06 15:02:54	USBSTOR and IEUser Explorer keys	Removable-storage and Explorer registry timestamps
2019-01-06 15:03:06	MountPoints2 `F:` key	Mounted-volume-related registry timestamp
2019-01-06 15:03:07	MountPoints2 volume-GUID key	Related volume registry timestamp
2019-01-06 15:04:30–15:06:23	Multiple Chrome processes	Browser processes present
Approximately 2019-01-06 15:06:11	`cmd.exe`, PID 3996, and `conhost.exe`, PID 3988	Console processes associated in context with recovered command history
2019-01-06 15:07:15	`FTK Imager.exe`, PID 2596	Forensic imaging application process observed
2019-01-06 15:07:15	IEUser profile key	User-profile registry timestamp
2019-01-06 15:08:32	Chrome UDP endpoint	Network endpoint creation timestamp
2019-01-06 15:08:42	UserAssist Count key	UserAssist registry timestamp
2019-01-06 15:09:06	Imageinfo	Reported image timestamp
2019-01-06 15:09:07	`dllhost.exe`, PID 2228	Process start time one second after reported image time

The console history records the commands `ipconfig`, `cd Downloads`, and `copy secret_file.docx F:`. The saved output does not provide a separately verified timestamp for each command, so the commands should not be assigned a more precise time than the evidence supports.

The timeline also contains inconsistent process timestamps. Consequently, it should be treated as a correlation aid rather than a definitive sequence of every event that occurred.

13. KEY FINDINGS

Finding 1: Working-copy integrity

Observation: The original and working memory images produced the same SHA-256 digest and were recorded as the same size.

Interpretation: The analysis working copy was identical to the source at the time of verification.

Confidence: High for the recorded integrity comparison.

Finding 2: Windows 7 SP1 x86 environment

Observation: Imageinfo suggested Windows 7 SP1 x86 profiles, and the Windows version registry identified Windows 7 Enterprise, build 7601, Service Pack 1.

Interpretation: The memory image is consistent with a Windows 7 Enterprise SP1 x86 system.

Confidence: Moderate to high, subject to the limitations of profile selection.

Finding 3: Console history records a file copy command

Observation: Recovered console history contains `copy secret_file.docx F:` followed by `1 file(s) copied.`

Interpretation: The recovered console artifact is consistent with a reported successful copy operation.

Confidence: High that the command and success text are present in the recovered artifacts. The destination file itself was not independently verified.

Finding 4: F: is associated with USB storage

Observation: USBSTOR identifies a General UDisk device; MountedDevices maps `F:` to a USB storage path; MountPoints2 contains related entries.

Interpretation: The registry evidence is consistent with `F:` being associated with a USB storage volume.

Confidence: Moderate to high for the registry association. Physical ownership and authorisation remain unknown.

Finding 5: Network artifacts were recovered

Observation: The image contains a host address of `192.168.56.8`, two closed TCP records involving `192.168.56.5:80`, a Chrome UDP endpoint, and an SSH listener on port 22.

Interpretation: Network related structures were present in the memory image.

Confidence: High for the presence of the recorded structures. The purpose and contents of the network activity were not established.

Finding 6: UserAssist and RecentDocs contain application related labels

Observation: RecentDocs and manually decoded UserAssist strings include script names and executable paths.

Interpretation: These artifacts provide context about recorded user activity traces but do not prove that every referenced file was executed.

Confidence: Moderate for the presence of the strings; limited for any conclusion about execution or intent.

Finding 7: Timestamp inconsistencies and plugin limitations affect interpretation

Observation: Some process timestamps conflict with the image timestamp, `psscan` returned no records, and several plugin imports failed because dependencies were missing.

Interpretation: The examination could not conclusively resolve every process or user activity question.

Confidence: High that the recorded limitations exist; their underlying cause remains unestablished.

14. LIMITATIONS

The following limitations apply to the findings:

1. **Single evidence source:** The examination focused on one supplied memory image. No complete disk image or independent USB-device image was examined.
2. **Destination file not verified:** The existence, content, integrity, and persistence of the destination copy of `secret_file.docx` were not independently established.
3. **Plugin dependencies:** Some Volatility plugins failed because `Crypto.Hash` was unavailable. Other plugins requiring `distorm3` were also unavailable.
4. **Process-scan result:** `psscan` returned no records. This does not establish that hidden or terminated processes were absent.

5. **Timestamp anomalies:** Some process times conflicted with other timestamps in the image. The cause was not established.
6. **Network attribution:** Closed TCP records with PID `-1` do not prove successful data transfer or identify the content or purpose of a connection.
7. **User-activity interpretation:** RecentDocs and UserAssist artifacts do not independently prove successful execution or malicious intent.
8. **Device attribution:** The USB storage description does not identify its physical owner or establish whether its use was authorised.
9. **Source provenance:** Matching hashes demonstrate consistency between the source and working copy during verification, not the authenticity of the source before receipt.

15. CONCLUSION

The examination of `memdumpWin7.mem` recovered process, network, console, and registry artifacts consistent with a Windows 7 Enterprise SP1 x86 environment.

The strongest investigative finding is the relationship between the recovered console command:

`copy secret_file.docx F:`

and the console response:

`1 file(s) copied.`

The registry artifacts independently identify a General UDisk USB storage device and associate drive letter `F:` with a USB storage path. Considered together, these artifacts support the inference that the captured system retained evidence consistent with copying a file named `secret_file.docx` to a removable storage volume.

The evidence does not independently establish the file's contents, the identity of the person who performed the operation, whether the action was authorised, or whether the file was subsequently transferred elsewhere.

The investigation also identified network related artifacts, an SSH listening process, Chrome processes, an FTK Imager process, and user profile traces. These provide context but do not independently establish malicious activity.

The findings should be interpreted alongside the documented limitations, especially the missing plugin dependencies, empty `psscan` output, and timestamp inconsistencies.

16. RECOMMENDATIONS

The following further steps would help validate and extend the findings:

1. If available and authorised, acquire a forensic image of the suspected USB device using an appropriate write blocked process.
2. Calculate and document cryptographic hashes for any recovered copy of `secret_file.docx`.
3. Compare the destination file with any source copy recovered from the system disk.
4. Examine relevant disk artifacts, including LNK files, Prefetch, Amcache, event logs, and other file system metadata, where available.
5. If network activity is relevant, obtain independent packet captures, firewall logs, proxy records, or server logs relating to `192.168.56.5:80`.
6. Repeat selected memory analysis tasks with a compatible, dependency complete toolchain and independently validate the important findings.
7. Maintain the original evidence, preserve hashes, and document all additional evidence handling actions.

17. COMMANDS USED

The following commands represent the principal commands recorded in the workflow. The complete command history and saved output files should be retained with the case materials. Some exploratory commands and focused registry queries are represented by their corresponding output filenames below.

17.1 Evidence integrity and file metadata

```bash
cd ~/SBT-DF204-CaseStudy3

sha256sum original/memdumpWin7.mem
sha256sum working/memdumpWin7.mem

stat -c '%n | %s bytes | permissions: %A' \
  original/memdumpWin7.mem \
  working/memdumpWin7.mem

file original/memdumpWin7.mem working/memdumpWin7.mem

find outputs evidence -maxdepth 1 -type f \
  -printf '%p | %s bytes\n' | sort
```

17.2 Volatility installation and image identification

```bash
python2 ~/tools/volatility/vol.py --info

python2 ~/tools/volatility/vol.py \
  -f working/memdumpWin7.mem imageinfo 2>&1 \
  | tee outputs/01_imageinfo.txt
```

KDBG scan:

```bash
python2 ~/tools/volatility/vol.py \
  -f working/memdumpWin7.mem \
  --profile=Win7SP1x86 kdbgscan 2>&1 \
  | tee outputs/04_kdbgscan.txt
```

17.3 Process examination

```bash
python2 ~/tools/volatility/vol.py \
  -f working/memdumpWin7.mem \
  --profile=Win7SP1x86 pslist 2>&1 \
  | tee outputs/02_pslist_Win7SP1x86.txt

python2 ~/tools/volatility/vol.py \
  -f working/memdumpWin7.mem \
  --profile=Win7SP1x86 psscan 2>&1 \
  | tee outputs/03_psscan_Win7SP1x86.txt
```

17.4 Network and console examination

```bash
python2 ~/tools/volatility/vol.py \
  -f working/memdumpWin7.mem \
  --profile=Win7SP1x86 netscan 2>&1 \
  | tee outputs/05_netscan.txt

python2 ~/tools/volatility/vol.py \
  -f working/memdumpWin7.mem \
  --profile=Win7SP1x86 cmdscan 2>&1 \
  | tee outputs/06_cmdscan.txt

python2 ~/tools/volatility/vol.py \
  -f working/memdumpWin7.mem \
  --profile=Win7SP1x86 consoles 2>&1 \
  | tee outputs/07_consoles.txt
```

17.5 Registry hive enumeration

```bash
python2 ~/tools/volatility/vol.py \
  -f working/memdumpWin7.mem \
  --profile=Win7SP1x86 hivelist 2>&1 \
  | tee outputs/08_hivelist.txt
```

SYSTEM `Select` key:

```bash
python2 ~/tools/volatility/vol.py \
  -f working/memdumpWin7.mem \
  --profile=Win7SP1x86 \
  printkey -K "Select" -o 0x87c1c008 2>&1 \
  | tee outputs/09_system_select.txt
```

USBSTOR query:

```bash
python2 ~/tools/volatility/vol.py \
  -f working/memdumpWin7.mem \
  --profile=Win7SP1x86 \
  printkey -K "Enum\\USBSTOR" -o 0x87c1c008 2>&1 \
  | tee outputs/10_usbstor_controlset001.txt
```

MountedDevices query:

```bash
python2 ~/tools/volatility/vol.py \
  -f working/memdumpWin7.mem \
  --profile=Win7SP1x86 \
  printkey -K "MountedDevices" -o 0x87c1c008 2>&1 \
  | tee outputs/12_mounted_devices.txt
```

```

**Windows version query:**

```bash
python2 ~/tools/volatility/vol.py \
  -f working/memdumpWin7.mem \
  --profile=Win7SP1x86 \
  printkey -K "Microsoft\\Windows NT\\CurrentVersion" \
  -o 0x9711d9c8 2>&1 \
  | tee outputs/15_windows_version.txt
```

Additional focused queries examined the USB device instance, ProfileList SIDs, IEUser Explorer key, Run key, RecentDocs, TypedPaths, RunMRU, MountPoints2, and UserAssist Count key.

The corresponding saved output files are listed in the next section. The exact hive offsets and key paths for repeat execution should be taken from the saved command history and original output files.

18. OUTPUT AND EVIDENCE INVENTORY

The following saved files formed part of the analysis record.

File	Purpose
`evidence/01_source_metadata.txt`	Source file metadata
`evidence/02_copy_verification.txt`	Source and working-copy verification
`evidence/03_tool_inventory.txt`	Tool inventory
`outputs/01_imageinfo.txt`	Memory-image identification and profile suggestions
`outputs/02_pslist_Win7SP1x86.txt`	Process listing
`outputs/03_psscan_Win7SP1x86.txt`	Process-scan result
`outputs/04_kdbgscan.txt`	KDBG scan
`outputs/05_netscan.txt`	Network artifacts
`outputs/06_cmdscan.txt`	Recovered command history
`outputs/07_consoles.txt`	Recovered console output
`outputs/08_hivelist.txt`	Registry hive list
`outputs/09_system_select.txt`	SYSTEM Select key
`outputs/10_usbstor_controlset001.txt`	USBSTOR ControlSet001
`outputs/11_usbstor_controlset002.txt`	USBSTOR ControlSet002
`outputs/12_mounted_devices.txt`	MountedDevices mapping
`outputs/13_usb_device_details.txt`	USB device details
`outputs/15_windows_version.txt`	Windows version registry output
`outputs/16_profile_list.txt`	User profile list
`outputs/17_usb_instance_corrected.txt`	Corrected USB instance query
`outputs/18_profile_sid_1000.txt`	IEUser profile details
`outputs/19_profile_sid_1002.txt`	sshd_server profile details
`outputs/20_usb_instance_details.txt`	USB instance details
`outputs/21_pslist.txt`	Additional process listing
`outputs/22_psscan.txt`	Additional process-scan output
`outputs/23_ieuser_explorer_registry.txt`	IEUser Explorer key
`outputs/24_ieuser_run_key.txt`	IEUser Run key
`outputs/25_sshd_explorer_registry.txt`	sshd_server Explorer-key query
`outputs/26_ieuser_recentdocs.txt`	RecentDocs artifacts
`outputs/27_ieuser_typedpaths.txt`	TypedPaths query
`outputs/28_ieuser_runmru.txt`	RunMRU query
`outputs/29_ieuser_mountpoints2.txt`	MountPoints2 artifacts
`outputs/30_ieuser_userassist.txt`	Earlier UserAssist-related output
```

`outputs/30_mountpoints2_f.txt`	MountPoints2 F: query
`outputs/31_userassist_count_guid1.txt`	Raw UserAssist Count key output
`outputs/31_mountpoints2_usb_guid.txt`	MountPoints2 USB volume-GUID query

Some filenames use repeated numeric prefixes because outputs were generated during successive stages of the investigation. The original filenames should be preserved when submitting the evidence package.

19. SCREENSHOT AND EXHIBIT REGISTER

Exhibit	Description
`SS01_Case_Folder_Structure.png`	Case workspace and directory layout
`SS02_Memory_Image_Location.png`	Source memory-image location
`SS03_Source_Comparison.png`	Source image comparison
`SS04_Working_Copy_Integrity.png`	Working-copy integrity verification
`SS05_Volatility_Installation_Verification.png`	Volatility installation and version
`SS06_Imageinfo_Profile_Candidates.png`	Imageinfo and suggested profiles
`SS07_Process_List_Profile_Test.png`	Process-list profile test
`SS08_Process_Scan_Profile_Test.png`	Process-scan profile test
`SS09_KDBG_Profile_Validation.png`	KDBG/profile validation
`SS10_System_Control_Set.png`	SYSTEM Select key
`SS11_USBSTOR_ControlSet001.png`	USBSTOR ControlSet001
`SS12_USBSTOR_ControlSet002.png`	USBSTOR ControlSet002
`SS13_Mounted_Drive_Mappings.png`	MountedDevices drive mappings
`SS14_Network_Scan.png`	Network scan output
`SS15_Command_History.png`	Recovered command history
`SS16_Console_History.png`	Recovered console transcript
`SS17_Registry_Hive_List.png`	Registry hive list
`SS18_USB_Device_Details.png`	USB device registry details
`SS19_Windows_Version.png`	Windows version and build
`SS20_User_Profile_List.png`	ProfileList results
`SS21_Process_List.png`	Primary process listing
`SS22_Process_Scan.png`	Primary process-scan output
`SS23_USB_Device_Details.png`	Additional USB instance details, if separately captured
`SS24_Network_Connections.png`	Network endpoint results; split into numbered parts if necessary
`SS25_IEUser_Explorer_Registry.png`	IEUser Explorer registry key
`SS26_IEUser_Run_Key.png`	IEUser Run key result
`SS27_SSHD_Explorer_Key_Result.png`	sshd_server Explorer-key query
`SS28_UserAssist_Count_GUID1.png`	UserAssist raw Count key output
`SS29_IEUser_RecentDocs.png`	RecentDocs results
`SS30_IEUser_TypedPaths.png`	TypedPaths query
`SS31_IEUser_RunMRU.png`	RunMRU query
`SS32_IEUser_MountPoints2.png`	MountPoints2 subkeys
`SS33_MountPoints2_F_Details.png`	MountPoints2 F: details
`SS34_MountPoints2_USB_GUID.png`	MountPoints2 volume-GUID details
`SS35_Final_SHA256_Integrity.png`	Matching original and working-copy hashes
`SS36_Image_Permissions_and_Size.png`	File size and read-only permissions
`SS37_Output_Evidence_Inventory.png`	Output and evidence inventory; split into parts if necessary

20. FINAL EVIDENCE INTEGRITY STATEMENT

The final verification recorded the following SHA-256 digest for both the original and working memory images:

`500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e`

Both files were reported as 1,073,676,288 bytes, with permissions recorded as `-r--r--r--`.

The output and evidence inventory was also generated, and screenshots were captured separately.

These results support the integrity of the working copy relative to the source image at the time of verification. They do not independently establish the provenance or authenticity of the source image before it was received.

EXAMINER DECLARATION

I prepared this report from the recorded forensic workflow and the available output artifacts associated with the supplied memory image. The report distinguishes observed artifacts from interpretations and identifies limitations where the available evidence does not support a stronger conclusion.

Prepared by: Charles Uke
Student ID: 2025/FWSD/11466
Course: SBT-DF204 – Computer Forensics Case Studies
Date: 16 October 2026

End of Report

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ cd ~/SBT-DF204-CaseStudy3

VOL="$HOME/tools/volatility/vol.py"
IMG="$HOME/SBT-DF204-CaseStudy3/working/memdumpWin7.mem"
PROFILE="Win7SP1x86"
IEUSER_HIVE="0x8b5d39b0"

python2 "$VOL" -f "$IMG" --profile="$PROFILE" \
  printkey -o "$IEUSER_HIVE" \
  -K 'Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist' \
  2>&1 | tee outputs/30_ieuser_userassist.txt
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
```

```
Legend: (S) = Stable   (V) = Volatile

----------------------------------

Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: UserAssist (S)
Last updated: 2015-09-21 09:21:16 UTC+0000

Subkeys:
  (S) {CEBFF5CD-ACE2-4F4F-9178-9926F41749EA}
  (S) {F4E57C4B-2036-45F0-A9AB-443BCFE33D9F}

Values:

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$
```

```
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: F (S)
Last updated: 2019-01-06 15:03:06 UTC+0000

Subkeys:

Values:
```

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ cd ~/SBT-DF204-CaseStudy3

VOL="$HOME/tools/volatility/vol.py"
IMG="$HOME/SBT-DF204-CaseStudy3/working/memdumpWin7.mem"
PROFILE="Win7SP1x86"
IEUSER_HIVE="0x8b5d39b0"

python2 "$VOL" -f "$IMG" --profile="$PROFILE" \
  printkey -o "$IEUSER_HIVE" \
  -K 'Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2\F' \
  2>&1 | tee outputs/30_mountpoints2_f.txt

python2 "$VOL" -f "$IMG" --profile="$PROFILE" \
  printkey -o "$IEUSER_HIVE" \
  -K 'Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2\{421e7e52-11dd-11e9-b3cf-08002709e15d}' \
  2>&1 | tee outputs/31_mountpoints2_usb_guid.txt
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
```

```
Values:
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Legend: (S) = Stable   (V) = Volatile

----------
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: {421e7e52-11dd-11e9-b3cf-08002709e15d} (S)
Last updated: 2019-01-06 15:03:07 UTC+0000
```

```
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: {421e7e52-11dd-11e9-b3cf-08002709e15d} (S)
Last updated: 2019-01-06 15:03:07 UTC+0000

Subkeys:
  (S) shell

Values:

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ cd ~/SBT-DF204-CaseStudy3

echo "═══ ORIGINAL IMAGE HASH ═══"
sha256sum original/memdumpWin7.mem

echo
echo "═══ WORKING IMAGE HASH ═══"
sha256sum working/memdumpWin7.mem

echo
echo "═══ IMAGE FILE DETAILS ═══"
stat -c '%n | %s bytes | permissions: %A' \
  original/memdumpWin7.mem working/memdumpWin7.mem

echo
echo "═══ OUTPUT FILE INVENTORY ═══"
find outputs evidence -maxdepth 1 -type f -printf '%p | %s bytes\n' | sort
═══ ORIGINAL IMAGE HASH ═══
500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e  original/memdumpWin7.mem

═══ WORKING IMAGE HASH ═══
500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e  working/memdumpWin7.mem

═══ IMAGE FILE DETAILS ═══
original/memdumpWin7.mem | 1073676288 bytes | permissions: -r--r--r--
working/memdumpWin7.mem | 1073676288 bytes | permissions: -r--r--r--

═══ OUTPUT FILE INVENTORY ═══
evidence/01_source_metadata.txt | 819 bytes
evidence/02_copy_verification.txt | 181 bytes
evidence/03_tool_inventory.txt | 256 bytes
```

```
═══ ORIGINAL IMAGE HASH ═══
500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e  original/memdumpWin7.mem

═══ WORKING IMAGE HASH ═══
500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e  working/memdumpWin7.mem

═══ IMAGE FILE DETAILS ═══
original/memdumpWin7.mem | 1073676288 bytes | permissions: -r--r--r--
working/memdumpWin7.mem | 1073676288 bytes | permissions: -r--r--r--

═══ OUTPUT FILE INVENTORY ═══
evidence/01_source_metadata.txt | 819 bytes
evidence/02_copy_verification.txt | 181 bytes
evidence/03_tool_inventory.txt | 256 bytes
outputs/01_imageinfo.txt | 3057 bytes
outputs/02_pslist_Win7SP1x86.txt | 10039 bytes
outputs/03_psscan_Win7SP1x86.txt | 2557 bytes
outputs/04_kdbgscan.txt | 8095 bytes
outputs/05_netscan.txt | 8312 bytes
outputs/06_cmdscan.txt | 3098 bytes
outputs/07_consoles.txt | 6388 bytes
outputs/08_hivelist.txt | 3289 bytes
outputs/09_system_select.txt | 2649 bytes
outputs/10_usbstor_controlset001.txt | 2541 bytes
outputs/11_usbstor_controlset002.txt | 2541 bytes
outputs/12_mounted_devices.txt | 9614 bytes
outputs/13_usb_device_details.txt | 2550 bytes
outputs/14_usb_device_instance.txt | 2411 bytes
outputs/15_windows_version.txt | 12076 bytes
outputs/16_profile_list.txt | 2914 bytes
outputs/17_usb_instance_corrected.txt | 2411 bytes
outputs/18_profile_sid_1000.txt | 3031 bytes
```

```
═══ OUTPUT FILE INVENTORY ═══
evidence/01_source_metadata.txt | 819 bytes
evidence/02_copy_verification.txt | 181 bytes
evidence/03_tool_inventory.txt | 256 bytes
outputs/01_imageinfo.txt | 3057 bytes
outputs/02_pslist_Win7SP1×86.txt | 10039 bytes
outputs/03_psscan_Win7SP1×86.txt | 2557 bytes
outputs/04_kdbgscan.txt | 8095 bytes
outputs/05_netscan.txt | 8312 bytes
outputs/06_cmdscan.txt | 3098 bytes
outputs/07_consoles.txt | 6388 bytes
outputs/08_hivelist.txt | 3289 bytes
outputs/09_system_select.txt | 2649 bytes
outputs/10_usbstor_controlset001.txt | 2541 bytes
outputs/11_usbstor_controlset002.txt | 2541 bytes
outputs/12_mounted_devices.txt | 9614 bytes
outputs/13_usb_device_details.txt | 2550 bytes
outputs/14_usb_device_instance.txt | 2411 bytes
outputs/15_windows_version.txt | 12076 bytes
outputs/16_profile_list.txt | 2914 bytes
outputs/17_usb_instance_corrected.txt | 2411 bytes
outputs/18_profile_sid_1000.txt | 3031 bytes
outputs/19_profile_sid_1002.txt | 2997 bytes
outputs/20_usb_instance_details.txt | 3448 bytes
outputs/21_pslist.txt | 10039 bytes
outputs/22_psscan.txt | 2557 bytes
outputs/23_ieuser_explorer_registry.txt | 3519 bytes
outputs/24_ieuser_run_key.txt | 2500 bytes
outputs/25_sshd_explorer_registry.txt | 2411 bytes
outputs/26_ieuser_recentdocs.txt | 13299 bytes
outputs/27_ieuser_typedpaths.txt | 2507 bytes
outputs/28_ieuser_runmru.txt | 2503 bytes
```

```
outputs/18_profile_sid_1000.txt | 3031 bytes
outputs/19_profile_sid_1002.txt | 2997 bytes
outputs/20_usb_instance_details.txt | 3448 bytes
outputs/21_pslist.txt | 10039 bytes
outputs/22_psscan.txt | 2557 bytes
outputs/23_ieuser_explorer_registry.txt | 3519 bytes
outputs/24_ieuser_run_key.txt | 2500 bytes
outputs/25_sshd_explorer_registry.txt | 2411 bytes
outputs/26_ieuser_recentdocs.txt | 13299 bytes
outputs/27_ieuser_typedpaths.txt | 2507 bytes
outputs/28_ieuser_runmru.txt | 2503 bytes
outputs/29_ieuser_mountpoints2.txt | 2752 bytes
outputs/30_ieuser_userassist.txt | 2597 bytes
outputs/30_mountpoints2_f.txt | 2498 bytes
outputs/31_mountpoints2_usb_guid.txt | 2547 bytes
outputs/31_userassist_count_guid1.txt | 31652 bytes

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$
```

mouse pointer outside or press Ctrl+Alt.

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ cd ~/SBT-DF204-CaseStudy3

VOL="$HOME/tools/volatility/vol.py"
IMG="$HOME/SBT-DF204-CaseStudy3/working/memdumpWin7.mem"
PROFILE="Win7SP1x86"
IEUSER_HIVE="0x8b5d39b0"

echo "===== IEUSER EXPLORER USERACTIVITY REGISTRY ====="
python2 "$VOL" -f "$IMG" --profile="$PROFILE" \
  printkey -o "$IEUSER_HIVE" \
  -K 'Software\Microsoft\Windows\CurrentVersion\Explorer' \
  2>&1 | tee outputs/23_ieuser_explorer_registry.txt
===== IEUSER EXPLORER USERACTIVITY REGISTRY =====
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
```

```
Legend: (S) = Stable   (V) = Volatile

----------------------------------
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: Explorer (S)
Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:
  (S) Advanced
  (S) ApplicationDestinations
  (S) AutoplayHandlers
  (S) BitBucket
  (S) CabinetState
  (S) CD Burning
  (S) CIDSave
  (S) CLSID
  (S) ComDlg32
  (S) Discardable
  (S) FileExts
  (S) LowRegistry
  (S) MenuOrder
  (S) Modules
  (S) MountPoints2
  (S) NewShortcutHandlers
  (S) RecentDocs
  (S) RunMRU
  (S) SearchPlatform
  (S) Shell Folders
  (S) StartPage
  (S) StartPage2
  (S) Streams
  (S) StuckRects2
```

```
Values:
REG_DWORD     ExplorerStartupTraceRecorded : (S) 1
REG_BINARY    ShellState      : (S)
0×00000000  24 00 00 00 32 28 00 00 00 00 00 00 00 00 00 00   $...2(..........
0×00000010  00 00 00 00 01 00 00 00 12 00 00 00 00 00 00 00   ................
0×00000020  22 00 00 00                                       "...
REG_DWORD     CleanShutdown   : (S) 0
REG_DWORD     Browse For Folder Width : (S) 318
REG_DWORD     Browse For Folder Height : (S) 288
```

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ echo "===== IEUSER RUN KEY ====="
python2 "$VOL" -f "$IMG" --profile="$PROFILE" \
  printkey -o "$IEUSER_HIVE" \
  -K 'Software\Microsoft\Windows\CurrentVersion\Run' \
  2>&1 | tee outputs/24_ieuser_run_key.txt
===== IEUSER RUN KEY =====
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
```

```
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: Run (S)
Last updated: 2015-09-21 09:21:20 UTC+0000

Subkeys:

Values:
```

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ SSHD_HIVE="0×a65269c8"

echo "═══ SSHD SERVER EXPLORER REGISTRY ═══"
python2 "$VOL" -f "$IMG" --profile="$PROFILE" \
  printkey -o "$SSHD_HIVE" \
  -K 'Software\Microsoft\Windows\CurrentVersion\Explorer' \
  2>&1 | tee outputs/25_sshd_explorer_registry.txt
═══ SSHD SERVER EXPLORER REGISTRY ═══
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
```

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ cd ~/SBT-DF204-CaseStudy3

VOL="$HOME/tools/volatility/vol.py"
IMG="$HOME/SBT-DF204-CaseStudy3/working/memdumpWin7.mem"
PROFILE="Win7SP1x86"
IEUSER_HIVE="0x8b5d39b0"

echo "===== IEUSER RECENT DOCUMENTS ====="
python2 "$VOL" -f "$IMG" --profile="$PROFILE" \
  printkey -o "$IEUSER_HIVE" \
  -K 'Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs' \
  2>&1 | tee outputs/26_ieuser_recentdocs.txt
===== IEUSER RECENT DOCUMENTS =====
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
```

```
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: RecentDocs (S)
Last updated: 2019-01-06 14:32:14 UTC+0000

Subkeys:
  (S) .PS1
  (S) .txt
  (S) Folder

Values:
REG_BINARY    MRUListEx       : (S)
0x00000000  0c 00 00 00 05 00 00 00 04 00 00 00 03 00 00 00   ................
0x00000010  0b 00 00 00 0a 00 00 00 09 00 00 00 08 00 00 00   ................
0x00000020  07 00 00 00 01 00 00 00 06 00 00 00 00 00 00 00   ................
0x00000030  02 00 00 00 ff ff ff ff                           ........
REG_BINARY    2               : (S)
0x00000000  70 00 72 00 65 00 70 00 72 00 6f 00 76 00 69 00   p.r.e.p.r.o.v.i.
0x00000010  73 00 69 00 6f 00 6e 00 65 00 72 00 2e 00 70 00   s.i.o.n.e.r...p.
0x00000020  73 00 31 00 00 00 80 00 32 00 00 00 00 00 00 00   s.1.....2.......
0x00000030  00 00 00 00 70 72 65 70 72 6f 76 69 73 69 6f 6e   ....preprovision
0x00000040  65 72 2e 70 73 31 2e 6c 6e 6b 00 00 5a 00 08 00   er.ps1.lnk..Z...
0x00000050  04 00 ef be 00 00 00 00 00 00 00 00 2a 00 00 00   ............*...
0x00000060  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0x00000070  00 00 00 00 00 00 70 00 72 00 65 00 70 00 72 00   ......p.r.e.p.r.
0x00000080  6f 00 76 00 69 00 73 00 69 00 6f 00 6e 00 65 00   o.v.i.s.i.o.n.e.
0x00000090  72 00 2e 00 70 00 73 00 31 00 2e 00 6c 00 6e 00   r...p.s.1...l.n.
```

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ cd ~/SBT-DF204-CaseStudy3

VOL="$HOME/tools/volatility/vol.py"
IMG="$HOME/SBT-DF204-CaseStudy3/working/memdumpWin7.mem"
PROFILE="Win7SP1x86"
IEUSER_HIVE="0x8b5d39b0"

python2 "$VOL" -f "$IMG" --profile="$PROFILE" \
  printkey -o "$IEUSER_HIVE" \
  -K 'Software\Microsoft\Windows\CurrentVersion\Explorer\UserAssist\{CEBFF5CD-ACE2-4F4F-9178-9926F41749EA}\Count' \
  2>&1 | tee outputs/31_userassist_count_guid1.txt
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
```

```
Legend: (S) = Stable   (V) = Volatile

----------------------------------

Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: Count (S)
Last updated: 2019-01-06 15:08:42 UTC+0000

Subkeys:

Values:
REG_BINARY    Zvpebfbsg.Jvaqbjf.TrggvatFgnegrq : (S)
0×00000000  00 00 00 00 0e 00 00 00 15 00 00 00 a0 68 06 00   .............h..
0×00000010  00 00 80 bf 00 00 80 bf 00 00 80 bf 00 00 80 bf   ................
0×00000020  00 00 80 bf 00 00 80 bf 00 00 80 bf 00 00 80 bf   ................
0×00000030  00 00 80 bf 00 00 80 bf ff ff ff ff 72 cf 42 9e   ............r.B.
0×00000040  4e f4 d0 01 00 00 00 00                           N.......
REG_BINARY    HRZR_PGYFRFFVBA : (S)
0×00000000  00 00 00 00 a4 00 00 00 7f 01 00 00 61 c0 b1 00   ............a...
0×00000010  0e 00 00 00 15 00 00 00 a0 68 06 00 4d 00 69 00   .........h..M.i.
0×00000020  63 00 72 00 6f 00 73 00 6f 00 66 00 74 00 2e 00   c.r.o.s.o.f.t...
0×00000030  57 00 69 00 6e 00 64 00 6f 00 77 00 73 00 2e 00   W.i.n.d.o.w.s...
```

```
Legend: (S) = Stable   (V) = Volatile

----------------------------
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: RecentDocs (S)
Last updated: 2019-01-06 14:32:14 UTC+0000

Subkeys:
  (S) .PS1
  (S) .txt
  (S) Folder

Values:
REG_BINARY    MRUListEx       : (S)
0x00000000  0c 00 00 00 05 00 00 00 04 00 00 00 03 00 00 00   ................
0x00000010  0b 00 00 00 0a 00 00 00 09 00 00 00 08 00 00 00   ................
0x00000020  07 00 00 00 01 00 00 00 06 00 00 00 00 00 00 00   ................
0x00000030  02 00 00 00 ff ff ff ff                           ........
REG_BINARY    2               : (S)
0x00000000  70 00 72 00 65 00 70 00 72 00 6f 00 76 00 69 00   p.r.e.p.r.o.v.i.
0x00000010  73 00 69 00 6f 00 6e 00 65 00 72 00 2e 00 70 00   s.i.o.n.e.r...p.
0x00000020  73 00 31 00 00 00 80 00 32 00 00 00 00 00 00 00   s.1.....2.......
0x00000030  00 00 00 00 70 72 65 70 72 6f 76 69 73 69 6f 6e   ....preprovision
0x00000040  65 72 2e 70 73 31 2e 6c 6e 6b 00 00 5a 00 08 00   er.ps1.lnk..Z...
0x00000050  04 00 ef be 00 00 00 00 00 00 00 00 2a 00 00 00   ............*...
0x00000060  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00   ................
0x00000070  00 00 00 00 00 00 70 00 72 00 65 00 70 00 72 00   ......p.r.e.p.r.
0x00000080  6f 00 76 00 69 00 73 00 69 00 6f 00 6e 00 65 00   o.v.i.s.i.o.n.e.
```

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ cd ~/SBT-DF204-CaseStudy3

VOL="$HOME/tools/volatility/vol.py"
IMG="$HOME/SBT-DF204-CaseStudy3/working/memdumpWin7.mem"
PROFILE="Win7SP1x86"
IEUSER_HIVE="0x8b5d39b0"

python2 "$VOL" -f "$IMG" --profile="$PROFILE" \
  printkey -o "$IEUSER_HIVE" \
  -K 'Software\Microsoft\Windows\CurrentVersion\Explorer\RecentDocs' \
  2>&1 | tee outputs/26_ieuser_recentdocs.txt

python2 "$VOL" -f "$IMG" --profile="$PROFILE" \
  printkey -o "$IEUSER_HIVE" \
  -K 'Software\Microsoft\Windows\CurrentVersion\Explorer\TypedPaths' \
  2>&1 | tee outputs/27_ieuser_typedpaths.txt
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
```

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ echo "═══ IEUSER TYPED PATHS ═══"
python2 "$VOL" -f "$IMG" --profile="$PROFILE" \
  printkey -o "$IEUSER_HIVE" \
  -K 'Software\Microsoft\Windows\CurrentVersion\Explorer\TypedPaths' \
  2>&1 | tee outputs/27_ieuser_typedpaths.txt
═══ IEUSER TYPED PATHS ═══
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Legend: (S) = Stable   (V) = Volatile
```

```
Legend: (S) = Stable   (V) = Volatile

----------------------------------------

Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: TypedPaths (S)
Last updated: 2015-09-21 09:45:19 UTC+0000

Subkeys:

Values:

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$
```

```
echo "===== IEUSER RUN MRU ====="
python2 "$VOL" -f "$IMG" --profile="$PROFILE" \
  printkey -o "$IEUSER_HIVE" \
  -K 'Software\Microsoft\Windows\CurrentVersion\Explorer\RunMRU' \
  2>&1 | tee outputs/28_ieuser_runmru.txt

echo "===== IEUSER MOUNT POINTS ====="
python2 "$VOL" -f "$IMG" --profile="$PROFILE" \
  printkey -o "$IEUSER_HIVE" \
  -K 'Software\Microsoft\Windows\CurrentVersion\Explorer\MountPoints2' \
  2>&1 | tee outputs/29_ieuser_mountpoints2.txt
===== IEUSER RUN MRU =====
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
```

```
Legend: (S) = Stable   (V) = Volatile

----------------------------------------

Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: RunMRU (S)
Last updated: 2015-09-23 10:16:38 UTC+0000

Subkeys:

Values:
```

```
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Legend: (S) = Stable   (V) = Volatile

----------
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: TypedPaths (S)
Last updated: 2015-09-21 09:45:19 UTC+0000

Subkeys:
```

```
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: TypedPaths (S)
Last updated: 2015-09-21 09:45:19 UTC+0000

Subkeys:

Values:

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$
```

```
Legend: (S) = Stable   (V) = Volatile

----------------------------------

Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: MountPoints2 (S)
Last updated: 2019-01-06 15:03:07 UTC+0000

Subkeys:
  (S) CPC
  (S) F
  (S) {421e7e52-11dd-11e9-b3cf-08002709e15d}
  (S) {762f4ebc-60ea-11e5-83af-806e6f6e6963}
  (S) {8358fe6d-60fa-11e5-bb4a-806e6f6e6963}
  (S) {a5b8a980-608c-11e5-a266-806e6f6e6963}
  (S) {a5b8a983-608c-11e5-a266-806e6f6e6963}

Values:

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$
```

```
Values:
===== IEUSER MOUNT POINTS =====
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Legend: (S) = Stable   (V) = Volatile

----------
Registry: \??\C:\Users\IEUser\ntuser.dat
Key name: MountPoints2 (S)
Last updated: 2019-01-06 15:03:07 UTC+0000
```

```
**************************************************
Instantiating KDBG using: /home/kali/SBT-DF204-CaseStudy3/working/memdumpWin7.mem WinXPSP2x86 (5.1.0 32bit)
Offset (P)                    : 0x2972c68
KDBG owner tag check          : True
Profile suggestion (KDBGHeader): Win7SP0x86
Version64                     : 0x2972c40 (Major: 15, Minor: 7601)
PsActiveProcessHead           : 0x8298b4f0
PsLoadedModuleList            : 0x82992e30
KernelBase                    : 0x82847000

**************************************************
Instantiating KDBG using: /home/kali/SBT-DF204-CaseStudy3/working/memdumpWin7.mem WinXPSP2x86 (5.1.0 32bit)
Offset (P)                    : 0xd6c7c68
KDBG owner tag check          : True
Profile suggestion (KDBGHeader): Win7SP1x86_23418
Version64                     : 0xd6c7c40 (Major: 15, Minor: 7601)
PsActiveProcessHead           : 0x8298b4f0
PsLoadedModuleList            : 0x82992e30
KernelBase                    : 0x82847000

**************************************************
Instantiating KDBG using: /home/kali/SBT-DF204-CaseStudy3/working/memdumpWin7.mem WinXPSP2x86 (5.1.0 32bit)
Offset (P)                    : 0xd6c7c68
KDBG owner tag check          : True
Profile suggestion (KDBGHeader): Win7SP1x86_24000
Version64                     : 0xd6c7c40 (Major: 15, Minor: 7601)
PsActiveProcessHead           : 0x8298b4f0
PsLoadedModuleList            : 0x82992e30
KernelBase                    : 0x82847000

**************************************************
```

```
**************************************************
Instantiating KDBG using: /home/kali/SBT-DF204-CaseStudy3/working/memdumpWin7.mem WinXPSP2x86 (5.1.0 32bit)
Offset (P)                    : 0xd6c7c68
KDBG owner tag check          : True
Profile suggestion (KDBGHeader): Win7SP1x86
Version64                     : 0xd6c7c40 (Major: 15, Minor: 7601)
PsActiveProcessHead           : 0x8298b4f0
PsLoadedModuleList            : 0x82992e30
KernelBase                    : 0x82847000

**************************************************
Instantiating KDBG using: /home/kali/SBT-DF204-CaseStudy3/working/memdumpWin7.mem WinXPSP2x86 (5.1.0 32bit)
Offset (P)                    : 0xd6c7c68
KDBG owner tag check          : True
Profile suggestion (KDBGHeader): Win7SP0x86
Version64                     : 0xd6c7c40 (Major: 15, Minor: 7601)
PsActiveProcessHead           : 0x8298b4f0
PsLoadedModuleList            : 0x82992e30
KernelBase                    : 0x82847000

**************************************************
Instantiating KDBG using: /home/kali/SBT-DF204-CaseStudy3/working/memdumpWin7.mem WinXPSP2x86 (5.1.0 32bit)
Offset (P)                    : 0xec94c68
KDBG owner tag check          : True
Profile suggestion (KDBGHeader): Win7SP1x86_23418
Version64                     : 0xec94c40 (Major: 15, Minor: 7601)
PsActiveProcessHead           : 0x8298b4f0
PsLoadedModuleList            : 0x82992e30
KernelBase                    : 0x82847000

**************************************************
```

```
Instantiating KDBG using: /home/kali/SBT-DF204-CaseStudy3/working/memdumpWin7.mem WinXPSP2×86 (5.1.0 32bit)
Offset (P)                    : 0×ec94c68
KDBG owner tag check          : True
Profile suggestion (KDBGHeader): Win7SP1×86_24000
Version64                     : 0×ec94c40 (Major: 15, Minor: 7601)
PsActiveProcessHead           : 0×8298b4f0
PsLoadedModuleList            : 0×82992e30
KernelBase                    : 0×82847000

**************************************************
Instantiating KDBG using: /home/kali/SBT-DF204-CaseStudy3/working/memdumpWin7.mem WinXPSP2×86 (5.1.0 32bit)
Offset (P)                    : 0×ec94c68
KDBG owner tag check          : True
Profile suggestion (KDBGHeader): Win7SP1×86
Version64                     : 0×ec94c40 (Major: 15, Minor: 7601)
PsActiveProcessHead           : 0×8298b4f0
PsLoadedModuleList            : 0×82992e30
KernelBase                    : 0×82847000

**************************************************
Instantiating KDBG using: /home/kali/SBT-DF204-CaseStudy3/working/memdumpWin7.mem WinXPSP2×86 (5.1.0 32bit)
Offset (P)                    : 0×ec94c68
KDBG owner tag check          : True
Profile suggestion (KDBGHeader): Win7SP0×86
Version64                     : 0×ec94c40 (Major: 15, Minor: 7601)
PsActiveProcessHead           : 0×8298b4f0
PsLoadedModuleList            : 0×82992e30
KernelBase                    : 0×82847000


┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$
```

```
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
**************************************************
CommandProcess: conhost.exe Pid: 2000
CommandHistory: 0x320960 Application: sshd.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x54
Cmd #29 @ 0x3200c4: 3?4?2???2
Cmd #37 @ 0x3200c4: 3?4?2???2
**************************************************
CommandProcess: conhost.exe Pid: 3988
CommandHistory: 0x110448 Application: cmd.exe Flags: Allocated, Reset
CommandCount: 3 LastAdded: 2 LastDisplayed: 2
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0x5c
Cmd #0 @ 0x10dc40: ipconfig
Cmd #1 @ 0x107f90: cd Downloads
Cmd #2 @ 0x1147a0: copy secret_file.docx F:
Cmd #22 @ 0xff818488: ?
Cmd #25 @ 0xff818488: ?
Cmd #36 @ 0xe00c4: ?????
Cmd #37 @ 0x10cff0: ?????

**************************************************
CONSOLE HISTORY
```

```
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: Select (S)
Last updated: 2009-07-14 04:53:15 UTC+0000

Subkeys:

Values:
REG_DWORD     Current          : (S) 1
REG_DWORD     Default          : (S) 1
REG_DWORD     Failed           : (S) 0
REG_DWORD     LastKnownGood    : (S) 2
```

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ cd ~/SBT-DF204-CaseStudy3

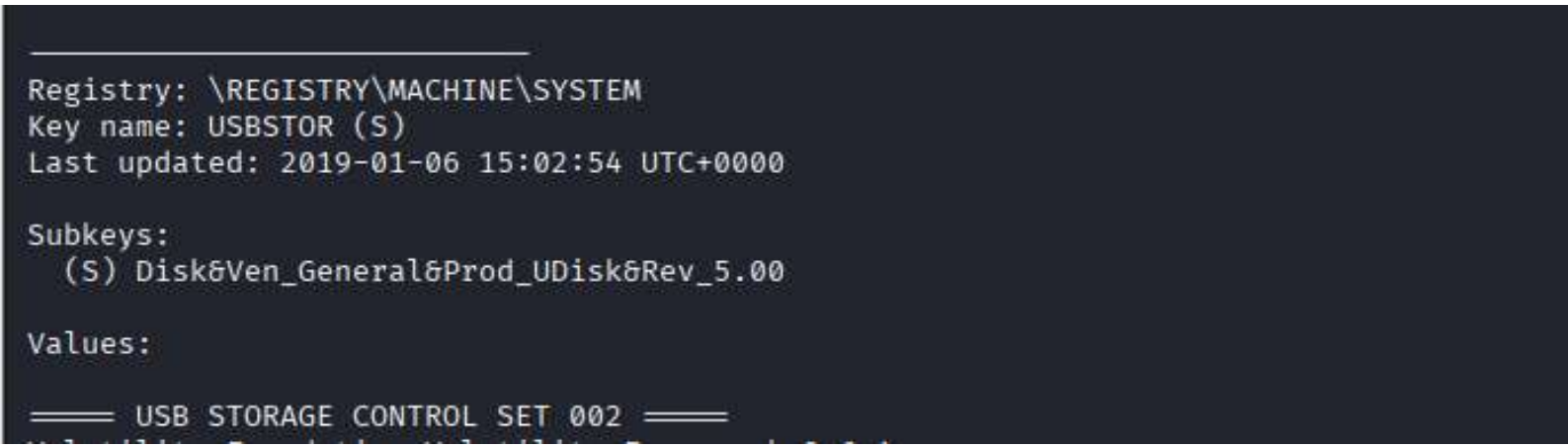
VOL="$HOME/tools/volatility/vol.py"
IMG="$HOME/SBT-DF204-CaseStudy3/working/memdumpWin7.mem"
PROFILE="Win7SP1x86"

echo "===== NETWORK CONNECTIONS ====="
python2 "$VOL" -f "$IMG" --profile="$PROFILE" netscan \
  2>&1 | tee outputs/05_netscan.txt

echo
echo "===== COMMAND HISTORY SCAN ====="
python2 "$VOL" -f "$IMG" --profile="$PROFILE" cmdscan \
  2>&1 | tee outputs/06_cmdscan.txt

echo
echo "===== CONSOLE HISTORY ====="
python2 "$VOL" -f "$IMG" --profile="$PROFILE" consoles \
  2>&1 | tee outputs/07_consoles.txt

echo
echo "===== REGISTRY HIVE LIST ====="
python2 "$VOL" -f "$IMG" --profile="$PROFILE" hivelist \
  2>&1 | tee outputs/08_hivelist.txt
===== NETWORK CONNECTIONS =====
```


Registry: \REGISTRY\MACHINE\SYSTEM
Key name: USBSTOR (S)
Last updated: 2019-01-06 15:02:54 UTC+0000
Subkeys:
(S) Disk&Ven_General&Prod_UDisk&Rev_5.00
Values:
USB STORAGE CONTROL SET 002

```
************************************************
ConsoleProcess: conhost.exe Pid: 2000
Console: 0×e181c0 CommandHistorySize: 50
HistoryBufferCount: 2 HistoryBufferMax: 4
OriginalTitle: C:\Program Files\OpenSSH\bin\cygrunsrv.exe
Title: C:\Program Files\OpenSSH\bin\cygrunsrv.exe
AttachedProcess: sshd.exe Pid: 2016 Handle: 0×54
----
CommandHistory: 0×320960 Application: sshd.exe Flags: Allocated
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0×54
----
CommandHistory: 0×3207f0 Application: - Flags:
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
ProcessHandle: 0×0
----
Screen 0×3360a0 X:80 Y:300
Dump:

************************************************
ConsoleProcess: conhost.exe Pid: 3988
Console: 0×e181c0 CommandHistorySize: 50
HistoryBufferCount: 2 HistoryBufferMax: 4
OriginalTitle: %SystemRoot%\system32\cmd.exe
Title: C:\Windows\system32\cmd.exe
AttachedProcess: cmd.exe Pid: 3996 Handle: 0×5c
----
CommandHistory: 0×1105d8 Application: ipconfig.exe Flags:
CommandCount: 0 LastAdded: -1 LastDisplayed: -1
FirstCommand: 0 CommandCountMax: 50
```

```
_____________________________________
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: USBSTOR (S)
Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:
  (S) Disk&Ven_General&Prod_UDisk&Rev_5.00

Values:

===== MOUNTED DRIVE MAPPINGS =====
```

```
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: MountedDevices (S)
Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:

Values:
REG_BINARY    \DosDevices\C:  : (S)
0×00000000  cd d7 ee 50 00 00 10 00 00 00 00 00                ...P........
REG_BINARY    \??\Volume{a5b8a980-608c-11e5-a266-806e6f6e6963} : (S)
0×00000000  cd d7 ee 50 00 00 10 00 00 00 00 00                ...P........
REG_BINARY    \DosDevices\D:  : (S)

REG_BINARY    \DosDevices\E:  : (S)
0×00000000  5c 00 3f 00 3f 00 5c 00 49 00 44 00 45 00 23 00   \.?.?.\.I.D.E.#.
0×00000010  43 00 64 00 52 00 6f 00 6d 00 56 00 42 00 4f 00   C.d.R.o.m.V.B.O.
0×00000020  58 00 5f 00 43 00 44 00 2d 00 52 00 4f 00 4d 00   X._.C.D.-.R.O.M.
0×00000030  5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00   _._._._._._._._.
0×00000040  5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00   _._._._._._._._.
0×00000050  5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00 5f 00   _._._._._._._._.
0×00000060  5f 00 5f 00 5f 00 5f 00 5f 00 31 00 2e 00 30 00   _._._._._.1...0.
0×00000070  5f 00 5f 00 5f 00 5f 00 5f 00 23 00 35 00 26 00   _._._._._.#.5.&.
0×00000080  33 00 39 00 34 00 63 00 30 00 61 00 64 00 33 00   3.9.4.c.0.a.d.3.
0×00000090  26 00 30 00 26 00 30 00 2e 00 31 00 2e 00 30 00   &.0.&.0...1...0.
0×000000a0  23 00 7b 00 35 00 33 00 66 00 35 00 36 00 33 00   #.{.5.3.f.5.6.3.
0×000000b0  30 00 64 00 2d 00 62 00 36 00 62 00 66 00 2d 00   0.d.-.b.6.b.f.-.
0×000000c0  31 00 31 00 64 00 30 00 2d 00 39 00 34 00 66 00   1.1.d.0.-.9.4.f.
0×000000d0  32 00 2d 00 30 00 30 00 61 00 30 00 63 00 39 00   2.-.0.0.a.0.c.9.
0×000000e0  31 00 65 00 66 00 62 00 38 00 62 00 7d 00         1.e.f.b.8.b.}.
REG_BINARY    \??\Volume{a5b8a985-608c-11e5-a266-806e6f6e6963} : (S)
0×00000000  5c 00 3f 00 3f 00 5c 00 46 00 44 00 43 00 23 00   \.?.?.\.F.D.C.#.
```

```
0×3e7f2828      UDPv4    0.0.0.0:0                *:*                                  1064    svchost.exe    2019-01-06 15:02:49
UTC+0000
0×3e7f2828      UDPv6    :::0                     *:*                                  1064    svchost.exe    2019-01-06 15:02:49
UTC+0000
0×3e446008      TCPv4    0.0.0.0:49154            0.0.0.0:0            LISTENING       880     svchost.exe
0×3e446d98      TCPv4    0.0.0.0:49154            0.0.0.0:0            LISTENING       880     svchost.exe
0×3e446d98      TCPv6    :::49154                 :::0                 LISTENING       880     svchost.exe
0×3e46e0c0      TCPv4    192.168.56.8:139         0.0.0.0:0            LISTENING       4       System
0×3e671988      TCPv4    0.0.0.0:445              0.0.0.0:0            LISTENING       4       System
0×3e671988      TCPv6    :::445                   :::0                 LISTENING       4       System
0×3e72e530      TCPv4    0.0.0.0:135              0.0.0.0:0            LISTENING       672     svchost.exe
0×3e72ee48      TCPv4    0.0.0.0:135              0.0.0.0:0            LISTENING       672     svchost.exe
0×3e72ee48      TCPv6    :::135                   :::0                 LISTENING       672     svchost.exe
0×3e736388      TCPv4    0.0.0.0:49152            0.0.0.0:0            LISTENING       340     wininit.exe
0×3e736d98      TCPv4    0.0.0.0:49152            0.0.0.0:0            LISTENING       340     wininit.exe
0×3e736d98      TCPv6    :::49152                 :::0                 LISTENING       340     wininit.exe
0×3e79d528      TCPv4    0.0.0.0:49153            0.0.0.0:0            LISTENING       724     svchost.exe
0×3e79d528      TCPv6    :::49153                 :::0                 LISTENING       724     svchost.exe
0×3e7a1618      TCPv4    0.0.0.0:49153            0.0.0.0:0            LISTENING       724     svchost.exe
0×3e043b10      TCPv4    192.168.56.8:49177       192.168.56.5:80      CLOSED          -1
0×3ec6c820      UDPv4    127.0.0.1:51123          *:*                                  2416    svchost.exe    2019-01-06 15:04:51
UTC+0000
0×3f19a820      UDPv6    ::1:1900                 *:*                                  2416    svchost.exe    2019-01-06 15:04:51
UTC+0000
0×3e8f0620      TCPv4    0.0.0.0:49156            0.0.0.0:0            LISTENING       440     lsass.exe
0×3e9b3e50      TCPv4    0.0.0.0:49155            0.0.0.0:0            LISTENING       432     services.exe
0×3e9b3e50      TCPv6    :::49155                 :::0                 LISTENING       432     services.exe
0×3ee1aa88      TCPv4    0.0.0.0:22               0.0.0.0:0            LISTENING       2016    sshd.exe
0×3ee1bc18      TCPv4    0.0.0.0:22               0.0.0.0:0            LISTENING       2016    sshd.exe
0×3ee1bc18      TCPv6    :::22                    :::0                 LISTENING       2016    sshd.exe
0×3fc0e008      UDPv4    127.0.0.1:1900           *:*                                  2416    svchost.exe    2019-01-06 15:04:51
UTC+0000
```

```
_____________________________
Registry: \REGISTRY\MACHINE\SYSTEM
Key name: Disk&Ven_General&Prod_UDisk&Rev_5.00 (S)
Last updated: 2019-01-06 15:02:54 UTC+0000

Subkeys:
  (S) 6&1bec0f48&0&_&0

Values:
```

```
===== WINDOWS VERSION =====
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Legend: (S) = Stable   (V) = Volatile

----------------------------
Registry: \SystemRoot\System32\Config\SOFTWARE
Key name: CurrentVersion (S)
Last updated: 2019-01-06 14:32:50 UTC+0000
```

```
Values:
REG_SZ        CurrentVersion  : (S) 6.1
REG_SZ        CurrentBuild    : (S) 7601
REG_SZ        SoftwareType    : (S) System
REG_SZ        CurrentType     : (S) Multiprocessor Free
REG_DWORD     InstallDate     : (S) 1442827050
REG_SZ        RegisteredOrganization : (S) Microsoft
REG_SZ        RegisteredOwner : (S)
REG_SZ        SystemRoot      : (S) C:\Windows
REG_SZ        InstallationType : (S) Client
REG_SZ        EditionID       : (S) Enterprise
REG_SZ        ProductName     : (S) Windows 7 Enterprise
REG_SZ        ProductId       : (S) 00392-972-8000024-85510
REG_BINARY    DigitalProductId : (S)
```

```
===== USER PROFILE LIST =====
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Legend: (S) = Stable   (V) = Volatile

----------
Registry: \SystemRoot\System32\Config\SOFTWARE
Key name: ProfileList (S)
Last updated: 2015-09-21 09:50:52 UTC+0000
```

```
Registry: \SystemRoot\System32\Config\SOFTWARE
Key name: ProfileList (S)
Last updated: 2015-09-21 09:50:52 UTC+0000

Subkeys:
  (S) S-1-5-18
  (S) S-1-5-19
  (S) S-1-5-20
  (S) S-1-5-21-1716914095-909560446-1177810406-1000
  (S) S-1-5-21-1716914095-909560446-1177810406-1002

Values:
REG_EXPAND_SZ ProfilesDirectory : (S) %SystemDrive%\Users
REG_EXPAND_SZ Default           : (S) %SystemDrive%\Users\Default
REG_EXPAND_SZ Public            : (S) %SystemDrive%\Users\Public
REG_EXPAND_SZ ProgramData       : (S) %SystemDrive%\ProgramData

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ cd ~/SBT-DF204-CaseStudy3

VOL="$HOME/tools/volatility/vol.py"
IMG="$HOME/SBT-DF204-CaseStudy3/working/memdumpWin7.mem"
PROFILE="Win7SP1x86"

echo "===== PROCESS LIST ====="
python2 "$VOL" -f "$IMG" --profile="$PROFILE" \
  pslist 2>&1 | tee outputs/21_pslist.txt

echo
echo "===== PROCESS SCAN ====="
python2 "$VOL" -f "$IMG" --profile="$PROFILE" \
  psscan 2>&1 | tee outputs/22_psscan.txt

echo
echo "===== NETWORK CONNECTIONS ====="
cat outputs/05_netscan.txt
===== PROCESS LIST =====
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
```

```
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Offset(V)  Name                    PID   PPID   Thds     Hnds   Sess  Wow64 Start                          Exit
---------- -------------------- ------ ------ ------ -------- ------ ------ ------------------------------ ------------------------------
0x8413a908 System                    4      0     73      518 ------      0 2019-01-06 18:02:44 UTC+0000
0x851ec488 smss.exe                236      4      2       29 ------      0 2019-01-06 18:02:44 UTC+0000
0x852b11e0 csrss.exe               300    292      8      434      0      0 2019-01-06 18:02:44 UTC+0000
0x841c32b8 wininit.exe             340    292      3       75      0      0 2019-01-06 18:02:44 UTC+0000
0x852b21d0 csrss.exe               348    328      9      316      1      0 2019-01-06 18:02:44 UTC+0000
0x858a4d28 winlogon.exe            388    328      3      112      1      0 2019-01-06 18:02:45 UTC+0000
0x858b0310 services.exe            432    340     10      228      0      0 2019-01-06 18:02:45 UTC+0000
0x858bbd28 lsass.exe               440    340      7      581      0      0 2019-01-06 18:02:45 UTC+0000
0x858be2c0 lsm.exe                 448    340     10      147      0      0 2019-01-06 18:02:45 UTC+0000
0x858f88d8 svchost.exe             552    432     10      359      0      0 2019-01-06 18:02:45 UTC+0000
0x8590cb00 VBoxService.ex          616    432     13      123      0      0 2019-01-06 18:02:45 UTC+0000
0x85913030 svchost.exe             672    432      8      260      0      0 2019-01-06 15:02:47 UTC+0000
0x85761030 svchost.exe             724    432     18      420      0      0 2019-01-06 15:02:47 UTC+0000
0x858d9208 svchost.exe             828    432     21      484      0      0 2019-01-06 15:02:47 UTC+0000
0x859d6518 svchost.exe             880    432     31      983      0      0 2019-01-06 15:02:47 UTC+0000
0x84f992c0 svchost.exe             992    432     13      275      0      0 2019-01-06 15:02:48 UTC+0000
0x85a23ce8 svchost.exe            1064    432     14      367      0      0 2019-01-06 15:02:48 UTC+0000
0x85a4fd28 spoolsv.exe            1184    432     12      272      0      0 2019-01-06 15:02:48 UTC+0000
0x85a91030 svchost.exe            1228    432     17      315      0      0 2019-01-06 15:02:48 UTC+0000
0x85ade488 vmicsvc.exe            1364    432      4       94      0      0 2019-01-06 15:02:48 UTC+0000
0x85ae3030 vmicsvc.exe            1388    432      5      105      0      0 2019-01-06 15:02:48 UTC+0000
0x85ae9030 taskhost.exe           1412    432      8      145      1      0 2019-01-06 15:02:48 UTC+0000
0x85af6d28 vmicsvc.exe            1440    432      3       66      0      0 2019-01-06 15:02:48 UTC+0000
0x85b06d28 vmicsvc.exe            1492    432      4       80      0      0 2019-01-06 15:02:48 UTC+0000
0x850553f0 vmicsvc.exe            1548    432      4       81      0      0 2019-01-06 15:02:48 UTC+0000
0x85b2dd28 svchost.exe            1588    432     10      147      0      0 2019-01-06 15:02:48 UTC+0000
0x859dc030 cygrunsrv.exe          1680    432      6      100      0      0 2019-01-06 15:02:48 UTC+0000
0x859a0d28 wlms.exe               1752    432      4       45      0      0 2019-01-06 15:02:48 UTC+0000
0x859bab00 taskeng.exe            1776    880      6       85      0      0 2019-01-06 15:02:48 UTC+0000
```

```
===== PROCESS SCAN =====
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Offset(P)          Name                PID   PPID PDB        Time created                   Time exited
------------------ ------------------ ------ ------ ---------- ------------------------------ ------------------------------

===== NETWORK CONNECTIONS =====
Volatility Foundation Volatility Framework 2.6.1
```

```
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Offset(V)  Name                    PID   PPID   Thds     Hnds   Sess  Wow64 Start                          Exit
---------- -------------------- ------ ------ ------ -------- ------ ------ ------------------------------ ------------------------------
0x8413a908 System                    4      0     73      518 ------      0 2019-01-06 18:02:44 UTC+0000
0x851ec488 smss.exe                236      4      2       29 ------      0 2019-01-06 18:02:44 UTC+0000
0x852b11e0 csrss.exe               300    292      8      434      0      0 2019-01-06 18:02:44 UTC+0000
0x841c32b8 wininit.exe             340    292      3       75      0      0 2019-01-06 18:02:44 UTC+0000
0x852b21d0 csrss.exe               348    328      9      316      1      0 2019-01-06 18:02:44 UTC+0000
0x858a4d28 winlogon.exe            388    328      3      112      1      0 2019-01-06 18:02:45 UTC+0000
0x858b0310 services.exe            432    340     10      228      0      0 2019-01-06 18:02:45 UTC+0000
0x858bbd28 lsass.exe               440    340      7      581      0      0 2019-01-06 18:02:45 UTC+0000
0x858be2c0 lsm.exe                 448    340     10      147      0      0 2019-01-06 18:02:45 UTC+0000
0x858f88d8 svchost.exe             552    432     10      359      0      0 2019-01-06 18:02:45 UTC+0000
0x8590cb00 VBoxService.ex          616    432     13      123      0      0 2019-01-06 18:02:45 UTC+0000
0x85913030 svchost.exe             672    432      8      260      0      0 2019-01-06 15:02:47 UTC+0000
0x85761030 svchost.exe             724    432     18      420      0      0 2019-01-06 15:02:47 UTC+0000
0x858d9208 svchost.exe             828    432     21      484      0      0 2019-01-06 15:02:47 UTC+0000
0x859d6518 svchost.exe             880    432     31      983      0      0 2019-01-06 15:02:47 UTC+0000
0x84f992c0 svchost.exe             992    432     13      275      0      0 2019-01-06 15:02:48 UTC+0000
0x85a23ce8 svchost.exe            1064    432     14      367      0      0 2019-01-06 15:02:48 UTC+0000
0x85a4fd28 spoolsv.exe            1184    432     12      272      0      0 2019-01-06 15:02:48 UTC+0000
0x85a91030 svchost.exe            1228    432     17      315      0      0 2019-01-06 15:02:48 UTC+0000
0x85ade488 vmicsvc.exe            1364    432      4       94      0      0 2019-01-06 15:02:48 UTC+0000
0x85ae3030 vmicsvc.exe            1388    432      5      105      0      0 2019-01-06 15:02:48 UTC+0000
0x85ae9030 taskhost.exe           1412    432      8      145      1      0 2019-01-06 15:02:48 UTC+0000
0x85af6d28 vmicsvc.exe            1440    432      3       66      0      0 2019-01-06 15:02:48 UTC+0000
0x85b06d28 vmicsvc.exe            1492    432      4       80      0      0 2019-01-06 15:02:48 UTC+0000
0x850553f0 vmicsvc.exe            1548    432      4       81      0      0 2019-01-06 15:02:48 UTC+0000
0x85b2dd28 svchost.exe            1588    432     10      147      0      0 2019-01-06 15:02:48 UTC+0000
0x859dc030 cygrunsrv.exe          1680    432      6      100      0      0 2019-01-06 15:02:48 UTC+0000
0x859a0d28 wlms.exe               1752    432      4       45      0      0 2019-01-06 15:02:48 UTC+0000
```

ouse pointer inside or press Ctrl+G

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ cd ~/SBT-DF204-CaseStudy3

VOL="$HOME/tools/volatility/vol.py"
IMG="$HOME/SBT-DF204-CaseStudy3/working/memdumpWin7.mem"
PROFILE="Win7SP1x86"
SYSTEM_HIVE="0x87c1c008"

echo "===== SYSTEM CONTROL SET SELECTION ====="
python2 "$VOL" -f "$IMG" --profile="$PROFILE" \
  printkey -o "$SYSTEM_HIVE" -K "Select" \
  2>&1 | tee outputs/09_system_select.txt

echo
echo "===== USB STORAGE CONTROL SET 001 ====="
python2 "$VOL" -f "$IMG" --profile="$PROFILE" \
  printkey -o "$SYSTEM_HIVE" -K "ControlSet001\Enum\USBSTOR" \
  2>&1 | tee outputs/10_usbstor_controlset001.txt

echo
echo "===== USB STORAGE CONTROL SET 002 ====="
python2 "$VOL" -f "$IMG" --profile="$PROFILE" \
  printkey -o "$SYSTEM_HIVE" -K "ControlSet002\Enum\USBSTOR" \
  2>&1 | tee outputs/11_usbstor_controlset002.txt

echo
echo "===== MOUNTED DRIVE MAPPINGS ====="
python2 "$VOL" -f "$IMG" --profile="$PROFILE" \
  printkey -o "$SYSTEM_HIVE" -K "MountedDevices" \
  2>&1 | tee outputs/12_mounted_devices.txt
===== SYSTEM CONTROL SET SELECTION =====
Volatility Foundation Volatility Framework 2.6.1
```

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$
cd ~/SBT-DF204-CaseStudy3

VOL="$HOME/tools/volatility/vol.py"
IMG="$HOME/SBT-DF204-CaseStudy3/working/memdumpWin7.mem"
PROFILE="Win7SP1x86"
SYSTEM_HIVE="0x87c1c008"

echo "===== USB DEVICE DETAILS ====="
python2 "$VOL" -f "$IMG" --profile="$PROFILE" \
  printkey -o "$SYSTEM_HIVE" \
  -K "ControlSet001\Enum\USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00" \
  2>&1 | tee outputs/13_usb_device_details.txt

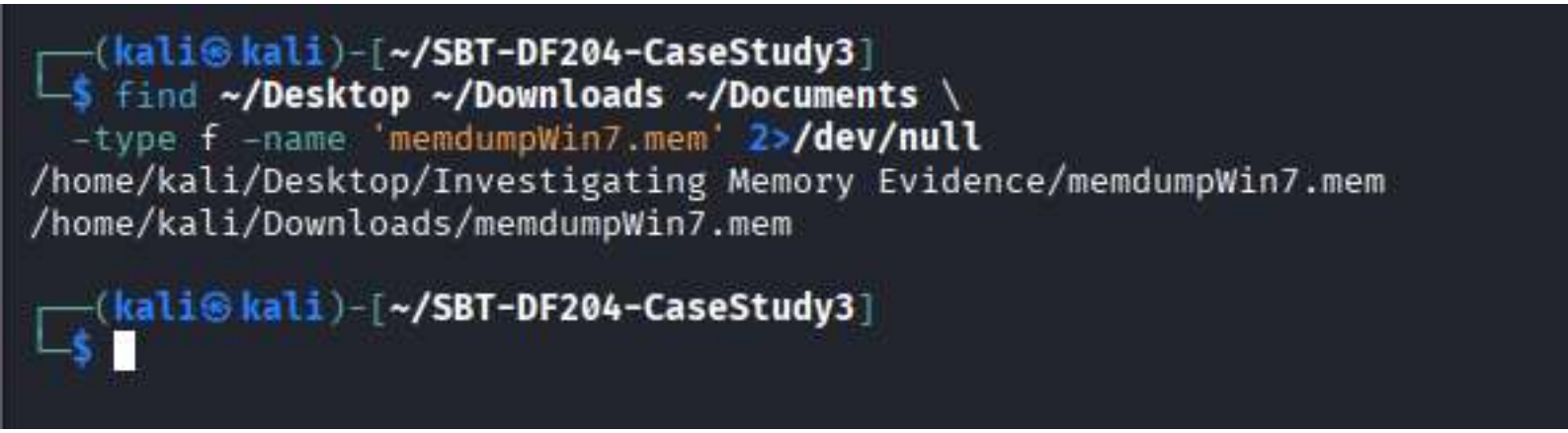
echo
echo "===== USB DEVICE INSTANCE DETAILS ====="
python2 "$VOL" -f "$IMG" --profile="$PROFILE" \
  printkey -o "$SYSTEM_HIVE" \
  -K "ControlSet001\Enum\USBSTOR\Disk&Ven_General&Prod_UDisk&Rev_5.00\6&1bec0f48&0&_\0" \
  2>&1 | tee outputs/14_usb_device_instance.txt

===== USB DEVICE DETAILS =====
```

```
zsh: corrupt history file /home/kali/.zsh_history
┌──(kali㉿kali)-[~]
└─$ mkdir -p ~/SBT-DF204-CaseStudy3/{original,working,evidence,outputs,screenshots,notes,report}
cd ~/SBT-DF204-CaseStudy3

pwd
ls -la
/home/kali/SBT-DF204-CaseStudy3
total 36
drwxrwxr-x  9 kali kali 4096 Oct 16 16:26 .
drwx------ 40 kali kali 4096 Oct 16 16:26 ..
drwxrwxr-x  2 kali kali 4096 Oct 16 16:26 evidence
drwxrwxr-x  2 kali kali 4096 Oct 16 16:26 notes
drwxrwxr-x  2 kali kali 4096 Oct 16 16:26 original
drwxrwxr-x  2 kali kali 4096 Oct 16 16:26 outputs
drwxrwxr-x  2 kali kali 4096 Oct 16 16:26 report
drwxrwxr-x  2 kali kali 4096 Oct 16 16:26 screenshots
drwxrwxr-x  2 kali kali 4096 Oct 16 16:26 working

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$
```


┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ find ~/Desktop ~/Downloads ~/Documents \
  -type f -name 'memdumpWin7.mem' 2>/dev/null
/home/kali/Desktop/Investigating Memory Evidence/memdumpWin7.mem
/home/kali/Downloads/memdumpWin7.mem
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ cp --preserve=mode,timestamps \
  original/memdumpWin7.mem \
  working/memdumpWin7.mem

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ sha256sum original/memdumpWin7.mem working/memdumpWin7.mem

stat -c '%n | %s bytes | permissions=%A' \
  original/memdumpWin7.mem \
  working/memdumpWin7.mem
500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e  original/memdumpWin7.mem
500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e  working/memdumpWin7.mem
original/memdumpWin7.mem | 1073676288 bytes | permissions=-r--r--r--
working/memdumpWin7.mem | 1073676288 bytes | permissions=-r--r--r--

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ sha256sum original/memdumpWin7.mem working/memdumpWin7.mem \
  | tee evidence/02_copy_verification.txt
500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e  original/memdumpWin7.mem
500bc1ba8b5ffa8053b6213d7e7aae0c84c18cce541dc53a55887e6e5099495e  working/memdumpWin7.mem

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ echo "═══ VOLATILITY COMMANDS ═══"
command -v volatility
command -v volatility2
command -v vol

echo
echo "═══ INSTALLED PACKAGES ═══"
dpkg -l | grep -i volatility

echo
echo "═══ COMMON INSTALLATION LOCATIONS ═══"
find /usr/share /usr/local /opt /home/kali/Desktop \
  -type f \( -name 'vol.py' -o -name 'volatility.py' \) \
  2>/dev/null | head -50

echo
echo "═══ POSSIBLE LAB ARCHIVES ═══"
find ~/Desktop ~/Downloads \
  -type f \( -iname '*volatility*.zip' -o -iname '*volatility*.tar.gz' \) \
  2>/dev/null
═══ VOLATILITY COMMANDS ═══

═══ INSTALLED PACKAGES ═══

═══ COMMON INSTALLATION LOCATIONS ═══

═══ POSSIBLE LAB ARCHIVES ═══

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ git --version
mkdir -p ~/tools
ls -la ~/tools
git version 2.50.1
total 8
drwxrwxr-x  2 kali kali 4096 Oct 16 16:55 .
drwx------ 41 kali kali 4096 Oct 16 16:55 ..

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ cd ~/tools
git clone https://github.com/volatilityfoundation/volatility.git
Cloning into 'volatility'...
remote: Enumerating objects: 27414, done.
remote: Counting objects: 100% (2/2), done.
remote: Compressing objects: 100% (2/2), done.
remote: Total 27414 (delta 0), reused 0 (delta 0), pack-reused 27412 (from 2)
Receiving objects: 100% (27414/27414), 21.11 MiB | 185.00 KiB/s, done.
Resolving deltas: 100% (19758/19758), done.

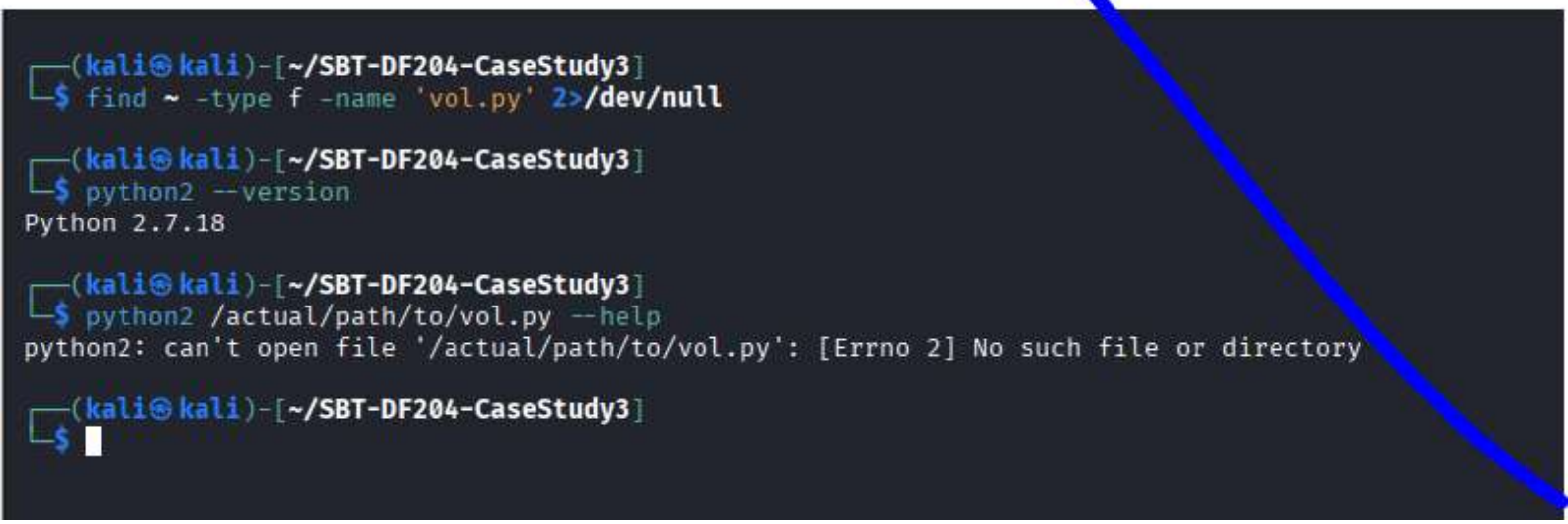
┌──(kali㉿kali)-[~/tools]
└─$ cd ~/tools/volatility

git rev-parse HEAD
python2 vol.py --help
328a178edeecaf8d2abf96618ecfcb14a7b0c860
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
```

```
┌──(kali㉿kali)-[~/tools]
└─$ cd ~/tools/volatility

git rev-parse HEAD
python2 vol.py --help
328a178edeecaf8d2abf96618ecfcb14a7b0c860
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Usage: Volatility - A memory forensics analysis platform.
```

```
Options:
  -h, --help            list all available options and their default values.
                        Default values may be set in the configuration file
                        (/etc/volatilityrc)
  --conf-file=/home/kali/.volatilityrc
                        User based configuration file
  -d, --debug           Debug volatility
  --plugins=PLUGINS     Additional plugin directories to use (colon separated)
  --info                Print information about all registered objects
  --cache-directory=/home/kali/.cache/volatility
                        Directory where cache files are stored
  --cache               Use caching
  --tz=TZ               Sets the (Olson) timezone for displaying timestamps
                        using pytz (if installed) or tzset
  -f FILENAME, --filename=FILENAME
                        Filename to use when opening an image
  --profile=WinXPSP2x86
                        Name of the profile to load (use --info to see a list
                        of supported profiles)
  -l LOCATION, --location=LOCATION
                        A URN location from which to load an address space
  -w, --write           Enable write support
  --dtb=DTB             DTB Address
  --shift=SHIFT         Mac KASLR shift address
  --physical_shift=PHYSICAL_SHIFT
                        Linux kernel physical shift address
  --virtual_shift=VIRTUAL_SHIFT
                        Linux kernel virtual shift address
  --output=text         Output in this format (support is module specific, see
                        the Module Output Options below)
  --output-file=OUTPUT_FILE
```


┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ find ~ -type f -name 'vol.py' 2>/dev/null
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ python2 --version
Python 2.7.18
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ python2 /actual/path/to/vol.py --help
python2: can't open file '/actual/path/to/vol.py': [Errno 2] No such file or directory
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$

```
┌──(kali㉿kali)-[~/tools/volatility]
└─$ cd ~/SBT-DF204-CaseStudy3

{
  echo "Tool: Volatility Framework 2"
  echo "Version: 2.6.1"
  echo "Python: Python 2.7.18"
  echo "Source revision: 328a178edeecaf8d2abf96618ecfcb14a7b0c860"
  echo "Script: /home/kali/tools/volatility/vol.py"
  echo
  echo "Dependency warnings observed:"
  echo "Crypto.Hash and distorm3 imports failed for some plugins."
} | tee evidence/03_tool_inventory.txt
Tool: Volatility Framework 2
Version: 2.6.1
Python: Python 2.7.18
Source revision: 328a178edeecaf8d2abf96618ecfcb14a7b0c860
Script: /home/kali/tools/volatility/vol.py

Dependency warnings observed:
Crypto.Hash and distorm3 imports failed for some plugins.
```

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ cd ~/SBT-DF204-CaseStudy3

python2 ~/tools/volatility/vol.py \
  -f working/memdumpWin7.mem \
  imageinfo \
  2>&1 | tee outputs/01_imageinfo.txt
Volatility Foundation Volatility Framework 2.6.1
INFO    : volatility.debug    : Determining profile based on KDBG search...
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
```

```
Session  Actions  Edit  View  Help
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
          Suggested Profile(s) : Win7SP1×86_23418, Win7SP0×86, Win7SP1×86_24000, Win7SP1×86
                     AS Layer1 : IA32PagedMemoryPae (Kernel AS)
                     AS Layer2 : FileAddressSpace (/home/kali/SBT-DF204-CaseStudy3/working/memdumpWin7.mem)
                      PAE type : PAE
                           DTB : 0×185000L
                          KDBG : 0×82972c68L
          Number of Processors : 1
     Image Type (Service Pack) : 1
                KPCR for CPU 0 : 0×82973d00L
             KUSER_SHARED_DATA : 0×ffdf0000L
           Image date and time : 2019-01-06 15:09:06 UTC+0000
     Image local date and time : 2019-01-06 07:09:06 -0800

┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$
```

```
┌──(kali㉿kali)-[~/SBT-DF204-CaseStudy3]
└─$ cd ~/SBT-DF204-CaseStudy3

VOL="$HOME/tools/volatility/vol.py"
IMG="$HOME/SBT-DF204-CaseStudy3/working/memdumpWin7.mem"
PROFILE="Win7SP1x86"

echo "===== PROCESS LIST: $PROFILE ====="
python2 "$VOL" -f "$IMG" --profile="$PROFILE" pslist \
  2>&1 | tee outputs/02_pslist_Win7SP1x86.txt

echo
echo "===== PROCESS SCAN: $PROFILE ====="
python2 "$VOL" -f "$IMG" --profile="$PROFILE" psscan \
  2>&1 | tee outputs/03_psscan_Win7SP1x86.txt

echo
echo "===== KERNEL DEBUGGER BLOCK SCAN ====="
python2 "$VOL" -f "$IMG" kdbgscan \
  2>&1 | tee outputs/04_kdbgscan.txt
===== PROCESS LIST: Win7SP1x86 =====
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
```

```
echo
echo "===== KERNEL DEBUGGER BLOCK SCAN ====="
python2 "$VOL" -f "$IMG" kdbgscan \
  2>&1 | tee outputs/04_kdbgscan.txt
===== PROCESS LIST: Win7SP1×86 =====
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Offset(V)  Name                    PID   PPID   Thds     Hnds   Sess  Wow64 Start                          Exit
```

```
0×8413a908 System                    4      0     73      518 ------      0 2019-01-06 18:02:44 UTC+0000
0×851ec488 smss.exe                236      4      2       29 ------      0 2019-01-06 18:02:44 UTC+0000
0×852b11e0 csrss.exe               300    292      8      434      0      0 2019-01-06 18:02:44 UTC+0000
0×841c32b8 wininit.exe             340    292      3       75      0      0 2019-01-06 18:02:44 UTC+0000
0×852b21d0 csrss.exe               348    328      9      316      1      0 2019-01-06 18:02:44 UTC+0000
0×858a4d28 winlogon.exe            388    328      3      112      1      0 2019-01-06 18:02:45 UTC+0000
0×858b0310 services.exe            432    340     10      228      0      0 2019-01-06 18:02:45 UTC+0000
0×858bbd28 lsass.exe               440    340      7      581      0      0 2019-01-06 18:02:45 UTC+0000
0×858be2c0 lsm.exe                 448    340     10      147      0      0 2019-01-06 18:02:45 UTC+0000
0×858f88d8 svchost.exe             552    432     10      359      0      0 2019-01-06 18:02:45 UTC+0000
0×8590cb00 VBoxService.ex          616    432     13      123      0      0 2019-01-06 18:02:45 UTC+0000
0×85913030 svchost.exe             672    432      8      260      0      0 2019-01-06 15:02:47 UTC+0000
0×85761030 svchost.exe             724    432     18      420      0      0 2019-01-06 15:02:47 UTC+0000
0×858d9208 svchost.exe             828    432     21      484      0      0 2019-01-06 15:02:47 UTC+0000
0×859d6518 svchost.exe             880    432     31      983      0      0 2019-01-06 15:02:47 UTC+0000
0×84f992c0 svchost.exe             992    432     13      275      0      0 2019-01-06 15:02:48 UTC+0000
0×85a23ce8 svchost.exe            1064    432     14      367      0      0 2019-01-06 15:02:48 UTC+0000
0×85a4fd28 spoolsv.exe            1184    432     12      272      0      0 2019-01-06 15:02:48 UTC+0000
0×85a91030 svchost.exe            1228    432     17      315      0      0 2019-01-06 15:02:48 UTC+0000
0×85ade488 vmicsvc.exe            1364    432      4       94      0      0 2019-01-06 15:02:48 UTC+0000
0×85ae3030 vmicsvc.exe            1388    432      5      105      0      0 2019-01-06 15:02:48 UTC+0000
0×85ae9030 taskhost.exe           1412    432      8      145      1      0 2019-01-06 15:02:48 UTC+0000
0×85af6d28 vmicsvc.exe            1440    432      3       66      0      0 2019-01-06 15:02:48 UTC+0000
0×85b06d28 vmicsvc.exe            1492    432      4       80      0      0 2019-01-06 15:02:48 UTC+0000
0×850553f0 vmicsvc.exe            1548    432      4       81      0      0 2019-01-06 15:02:48 UTC+0000
0×85b2dd28 svchost.exe            1588    432     10      147      0      0 2019-01-06 15:02:48 UTC+0000
0×859dc030 cygrunsrv.exe          1680    432      6      100      0      0 2019-01-06 15:02:48 UTC+0000
0×859a0d28 wlms.exe               1752    432      4       45      0      0 2019-01-06 15:02:48 UTC+0000
0×859bab00 taskeng.exe            1776    880      6       85      0      0 2019-01-06 15:02:48 UTC+0000
0×8500bd00 cygrunsrv.exe          1984   1680      0 ------      0      0 2019-01-06 15:02:49 UTC+0000   2019-01-06 15:02:49 UTC+0000
0×84202030 conhost.exe            2000    300      2       33      0      0 2019-01-06 15:02:49 UTC+0000
0×84203998 sshd.exe               2016   1984      4      100      0      0 2019-01-06 15:02:49 UTC+0000
```

```
0×8500bd00 cygrunsrv.exe          1984   1680      0 ————————      0      0 2019-01-06 15:02:49 UTC+0000   2019-01-06 15:02:49 UTC+0000
0×84202030 conhost.exe            2000    300      2       33      0      0 2019-01-06 15:02:49 UTC+0000
0×84203998 sshd.exe               2016   1984      4      100      0      0 2019-01-06 15:02:49 UTC+0000
0×8501cce8 GoogleUpdate.e          192   1776      4      109      0      0 2019-01-06 15:02:49 UTC+0000
0×85b11398 sppsvc.exe             1036    432      4      143      0      0 2019-01-06 15:02:50 UTC+0000
0×85a07030 svchost.exe            2072    432      5       91      0      0 2019-01-06 15:02:50 UTC+0000
0×857ce6d0 rundll32.exe           2320    552      4       85      1      0 2019-01-06 15:02:52 UTC+0000
0×84fddd28 dwm.exe                2512    828      3       70      1      0 2019-01-06 15:02:54 UTC+0000
0×85bba860 explorer.exe           2524   2500     32      952      1      0 2019-01-06 15:02:54 UTC+0000
0×85c7f030 VBoxTray.exe           2632   2524     13      140      1      0 2019-01-06 15:02:55 UTC+0000
0×85cb9820 SearchIndexer.         2960    432     13      607      0      0 2019-01-06 15:03:01 UTC+0000
0×85e287f0 WUDFHost.exe           3300    828      8      205      0      0 2019-01-06 15:03:07 UTC+0000
0×84fa2ad0 msiexec.exe            3748    432      4      174      0      0 2019-01-06 15:03:34 UTC+0000
0×841eb658 WmiPrvSE.exe           4032    552      5      114      0      0 2019-01-06 15:03:54 UTC+0000
0×8583a030 chrome.exe             2388   2524     33      819      1      0 2019-01-06 15:04:30 UTC+0000
0×851e8520 chrome.exe             1280   2388      7       82      1      0 2019-01-06 15:04:30 UTC+0000
0×850316b8 chrome.exe             2380   2388      2       57      1      0 2019-01-06 15:04:30 UTC+0000
0×852845c8 svchost.exe            2416    432     10      139      0      0 2019-01-06 15:04:30 UTC+0000
0×842b4d28 chrome.exe             2912   2388      9      159      1      0 2019-01-06 15:04:32 UTC+0000
0×858be710 svchost.exe            3784    432     14      370      0      0 2019-01-06 15:04:51 UTC+0000
0×8431fb08 cmd.exe                3996   2524      1       22      1      0 2019-01-06 15:06:11 UTC+0000
0×842ddd28 conhost.exe            3988    348      2       52      1      0 2019-01-06 15:06:11 UTC+0000
0×842eb6b0 chrome.exe              976   2388     14      198      1      0 2019-01-06 15:06:23 UTC+0000
0×84312030 FTK Imager.exe         2596   2524     14      358      1      0 2019-01-06 15:07:15 UTC+0000
0×850d5a38 dllhost.exe            2228    552      6 170 ... 38      1      0 2019-01-06 15:09:07 UTC+0000
```

```
===== PROCESS SCAN: Win7SP1×86 =====
Volatility Foundation Volatility Framework 2.6.1
*** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
*** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
*** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
*** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
Offset(P)          Name                PID   PPID PDB        Time created                   Time exited
------------------ ------------------ ----- ----- ---------- ------------------------------ ------------------------------
```

```
===== KERNEL DEBUGGER BLOCK SCAN =====
Volatility Foundation Volatility Framework 2.6.1
** Failed to import volatility.plugins.registry.shutdown (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.getservicesids (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.timeliner (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.malware.apihooks (NameError: name 'distorm3' is not defined)
** Failed to import volatility.plugins.malware.servicediff (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.registry.userassist (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.getsids (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.registry.shellbags (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.evtlogs (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.registry.shimcache (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.tcaudit (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.registry.dumpregistry (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.registry.lsadump (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.malware.threads (NameError: name 'distorm3' is not defined)
** Failed to import volatility.plugins.mac.apihooks_kernel (ImportError: No module named distorm3)
** Failed to import volatility.plugins.registry.amcache (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.mac.check_syscall_shadow (ImportError: No module named distorm3)
** Failed to import volatility.plugins.malware.svcscan (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.registry.auditpol (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.ssdt (NameError: name 'distorm3' is not defined)
** Failed to import volatility.plugins.registry.registryapi (ImportError: No module named Crypto.Hash)
** Failed to import volatility.plugins.mac.apihooks (ImportError: No module named distorm3)
** Failed to import volatility.plugins.envars (ImportError: No module named Crypto.Hash)
*********************************************
Instantiating KDBG using: /home/kali/SBT-DF204-CaseStudy3/working/memdumpWin7.mem WinXPSP2x86 (5.1.0 32bit)
Offset (P)                    : 0x2972c68
KDBG owner tag check          : True
Profile suggestion (KDBGHeader): Win7SP1x86_23418
Version64                     : 0x2972c40 (Major: 15, Minor: 7601)
```

```
Offset (P)                    : 0x2972c68
KDBG owner tag check          : True
Profile suggestion (KDBGHeader): Win7SP1x86_23418
Version64                     : 0x2972c40 (Major: 15, Minor: 7601)
PsActiveProcessHead           : 0x8298b4f0
PsLoadedModuleList            : 0x82992e30
KernelBase                    : 0x82847000

**************************************************
Instantiating KDBG using: /home/kali/SBT-DF204-CaseStudy3/working/memdumpWin7.mem WinXPSP2x86 (5.1.0 32bit)
Offset (P)                    : 0x2972c68
KDBG owner tag check          : True
Profile suggestion (KDBGHeader): Win7SP1x86_24000
Version64                     : 0x2972c40 (Major: 15, Minor: 7601)
PsActiveProcessHead           : 0x8298b4f0
PsLoadedModuleList            : 0x82992e30
KernelBase                    : 0x82847000

**************************************************
Instantiating KDBG using: /home/kali/SBT-DF204-CaseStudy3/working/memdumpWin7.mem WinXPSP2x86 (5.1.0 32bit)
Offset (P)                    : 0x2972c68
KDBG owner tag check          : True
Profile suggestion (KDBGHeader): Win7SP1x86
Version64                     : 0x2972c40 (Major: 15, Minor: 7601)
PsActiveProcessHead           : 0x8298b4f0
PsLoadedModuleList            : 0x82992e30
KernelBase                    : 0x82847000

**************************************************
Instantiating KDBG using: /home/kali/SBT-DF204-CaseStudy3/working/memdumpWin7.mem WinXPSP2x86 (5.1.0 32bit)
Offset (P)                    : 0x2972c68
KDBG owner tag check          : True
```